

Hierarchical conditional empirical Bayes matrix factorization

Gaussian, exponential and sparse priors for coupled molecular programs

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Abstract

We study multi-modal matrix factorization in which the distributions of sample loadings depend on earlier latent programs and fixed side information. A normalized directed prior links modality-specific additive factorizations, while hidden Markov priors regularize ordered feature factors. We derive a common inference construction for conditional Gaussian mixtures, exponential mixtures, optional fixed point masses, and the one- and two-slab CGB families. Each loading update combines a Gaussian normal-means likelihood, its own conditional prior, and the likelihood contributions of its children. Analytic component posteriors provide proposals for a Metropolis correction that preserves the joint latent target, including transitions to and from exact zero. We also derive augmented variational coordinates and complete-data parameter updates, distinguish exact identities from numerical approximations, and handle partially overlapping modalities by integrating or sampling missing latent programs. A comparison with population-prior matrix factorization establishes an explicit autoregressive representation of vector Gaussian mixtures and delineates the additional structural assumptions of the proposed framework. An experimental Gaussian-mixture implementation and independent numerical checks support the derivations; exponential-mixture solver support and comparative empirical validation remain future work. The construction does not imply causal identification or guaranteed improvement in either modality.

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1 Introduction

Multi-modal measurements can reveal different aspects of a sample’s latent biological programs. Separate matrix factorizations allow each modality to have its own programs, but independent priors do not express how the activation or amplitude of one program informs another. Conversely, substituting estimated loadings from one modality into a conditional prior for another treats those estimates as observed and omits feedback through their dependent programs.

Empirical Bayes matrix factorization reduces factor updates to normal-means problems [5], and cEBMF allows observed covariates to inform the prior [6]. TwinEB learns flexible population mixtures on entire row and column latent vectors [1]. Shared-subspace and hierarchical multi-omics models provide complementary integration strategies [7, 8, 15]. Building on these ideas, we consider normalized conditional loading distributions whose inputs include uncertain latent programs as well as fixed covariates. The inferential target is the resulting joint posterior, conditional on estimated parameters in an empirical-Bayes analysis.

The methodological construction has three parts. First, analytic normal-means posteriors for Gaussian, exponential and atomic components provide a common computational primitive. Second, child-prior contributions supply the missing upstream feedback, through either exact-target sampling or explicitly defined variational coordinates. Third, sequence updates for feature factors and missing-modality likelihoods complete the model without changing its directed loading-prior semantics. The resulting modularity covers the one- and two-slab CGB variants while retaining their actual support and scale constraints.

The paper develops these identities and their implementation requirements. It does not present a completed empirical benchmark. We distinguish the prototype’s implemented Gaussian/atomic sampler from derived extensions, and identify the experiments required to assess computational efficiency, uncertainty calibration and scientific utility. Section 14.2 positions the contribution relative to population-prior factorization; Section 14 also compares multi-omics alternatives.

2 Notation and assumptions

We observe paired cells in two matrices, with no requirement that their features or ranks agree:

$$Y^A \in \mathbb{R}^{N \times P_A}, \quad Y^R \in \mathbb{R}^{N \times P_R}, \quad A \in \mathbb{R}^{N \times K_A}, \quad R \in \mathbb{R}^{N \times K_R}, \quad F^m \in \mathbb{R}^{P_m \times K_m} \quad (m \in \{A, R\}).$$

Here $A = L_{\text{ATAC}}$, $R = L_{\text{RNA}}$, and $L^A = A$, $L^R = R$. The package stores factors in the orientation $L^m(F^m)^\top$. Indices i, t, k, j denote cells, features, ATAC components and RNA components, respectively. Ω_m is the set of observed entries. Rows are aligned between modalities; feature positions need not be aligned between modalities.

- (i) The observations have independent Gaussian errors conditional on latent variables and positive precisions τ_{it}^m . This matches the notebook’s Gaussian simulation. It does not assert a count likelihood for raw RNA or ATAC.
- (ii) Loading priors act within each cell, with parameters shared across cells. External covariates, if present, are fixed and are suppressed in most formulas.
- (iii) Any centering/scaling transformation supplied to a prior network is *fixed during the inference target*. Dynamic standardization is discussed in Section 12.1; it would otherwise invalidate the cell-local Markov blankets used below.
- (iv) Factor ordering, ranks, HMM state supports and component supports are fixed unless an explicitly evaluated model-selection step changes them.

- (v) The main derivation uses the unpenalized model, equivalent to a sparsity penalty of one. Optional regularization must be defined in the common objective (Section 12.2).

Write $\varphi(x; \mu, v)$ for a Gaussian density with *variance* $v > 0$. All distributions and log densities involving a spike are interpreted on their mixed discrete/continuous space, not by assigning an infinite ordinary density to zero. Empty products equal one; empty sums equal zero.

3 Hierarchical factorization model

3.1 Observation models

For $(i, t) \in \Omega_m$,

$$Y_{it}^m \mid L^m, F^m, \tau^m \sim \mathcal{N} \left(\sum_{k=1}^{K_m} L_{ik}^m F_{tk}^m, (\tau_{it}^m)^{-1} \right). \quad (3.1)$$

Denote the product likelihood by $p_m(Y^m \mid L^m, F^m, \tau^m)$. Unobserved entries contribute no likelihood term. Precisions may be known or shared within identifiable groups, such as a modality, a row or a column.

3.2 Independent and fixed-covariate fits

An independent baseline uses

$$p_0(A, R) = \prod_{i,k} g_k^A(A_{ik}) \prod_{i,j} g_j^R(R_{ij}). \quad (3.2)$$

A conditional RNA fit supplied with an estimated ATAC matrix $\hat{A}$ instead uses, for example,

$$p(R \mid \hat{A}) = \prod_i \prod_j g_j^R(R_{ij} \mid R_{i,<j}, \hat{A}_i). \quad (3.3)$$

This is a valid conditional model if its factors are normalized. Treating $\hat{A}$ as fixed does not infer ATAC loadings using RNA evidence. Even within (3.3), full RNA inference includes the effects of R_{ij} on later RNA priors. Plugging estimated earlier RNA loadings into successive EBNM calls does not include those effects.

3.3 Ordered joint loading prior

The proposed joint prior is

$$p(A, R \mid \theta) = \prod_{i=1}^N \left\{ \prod_{k=1}^{K_A} g_k^A(A_{ik} \mid A_{i,<k}; \theta_k^A) \prod_{j=1}^{K_R} g_j^R(R_{ij} \mid R_{i,<j}, A_i; \theta_j^R) \right\}. \quad (3.4)$$

It is normalized by integrating in reverse generative order. The within-cell ordering consists of all ATAC components followed by all RNA components. This is an autoregressive prior, not a causal identification result. Omitting unnecessary parent inputs gives a sparser directed graph.

For a generic loading node V_u , let $P_u = \text{pa}(V_u)$ and $\mathcal{C}_u = \text{ch}(V_u)$. With all proposed dependencies present,

$$\mathcal{C}_{A_{ik}} = \{A_{ih} : h > k\} \cup \{R_{ij} : 1 \leq j \leq K_R\}, \quad (3.5)$$

$$\mathcal{C}_{R_{ij}} = \{R_{ih} : h > j\}. \quad (3.6)$$

Only actual children contribute when a network structurally omits an input.

3.4 CGB loading prior and related families

The mathematical CGB family matching `cov_gb_prior.py` is

$$\pi_u(P_u) = \text{sigmoid}\{h_{\theta_u}(S_u(P_u))\}, \quad (3.7)$$

$$Z_u \mid P_u \sim \text{Bernoulli}(\pi_u(P_u)), \quad (3.8)$$

$$V_u \mid Z_u = 0, P_u = 0, \quad V_u \mid Z_u = 1, P_u \sim \mathcal{N}(\mu_u, d_u^2), \quad (3.9)$$

where S_u denotes a fixed affine preprocessing transformation and $d_u^2 > 0$. Parameters are shared over cells for each component. In the source, `pi_2` is our slab probability π_u , `pi_1` is $1 - \pi_u$, `mu_2` is μ_u , and `sigma2_sq` is d_u^2 . The slab is signed; CGB is neither an exact Bernoulli prior nor a positive truncated-normal prior. Only its mixing probability depends on parents.

More generally, replace the slab by a normalized distribution $G_u(dv \mid P_u)$:

$$g_u(dv \mid P_u) = (1 - \pi_u(P_u))\delta_0(dv) + \pi_u(P_u)G_u(dv \mid P_u). \quad (3.10)$$

Examples include a positive truncated Gaussian, a point-exponential model, or the strict binary slab $G_u = \delta_1$. The package's intended `gbinary` family uses a positive truncated Gaussian with $d_u = \omega|\mu_u|$; this is a different family from CGB. For such a slab, the normalizing constant is $\Phi(\mu_u/d_u)$, which reduces to $\Phi(1/\omega)$ only when $\mu_u > 0$. These are model equations, not a claim that every numerical boundary case of that separate solver is audited here.

3.5 HMM factor priors

For each modality and factor, independently across (m, k) , introduce a state chain $Q_{1:P_m, k}^m$ and mixture labels $C_{1:P_m, k}^m$. Suppress (m, k) :

$$Q_1 \sim \text{Categorical}(\alpha), \quad Q_t \mid Q_{t-1} = a \sim \text{Categorical}(T_{a, \cdot}), \quad (3.11)$$

$$C_t \mid Q_t = b \sim \text{Categorical}(\rho_{b, \cdot}), \quad F_t \mid Q_t = b, C_t = c \sim G_{bc}. \quad (3.12)$$

Thus

$$p_H(F, Q, C \mid \psi) = \alpha_{Q_1} \prod_{t=2}^{P_m} T_{Q_{t-1}, Q_t} \prod_{t=1}^{P_m} \rho_{Q_t, C_t} G_{Q_t, C_t}(dF_t). \quad (3.13)$$

Rows of T and ρ , and α , sum to one. The signed HMM uses $G_{bc} = \mathcal{N}(c_b, v_c)$ for $v_c > 0$, and $G_{bc} = \delta_{c_b}$ for $v_c = 0$. The positive model uses

$$G_{bc}^+(df) = \frac{\varphi(f; c_b, v_c) \mathbf{1}\{f > 0\}}{\Phi(c_b/\sqrt{v_c})} df \quad (v_c > 0), \quad (3.14)$$

with nonnegative atoms when $v_c = 0$. The negative model is the reflection of the positive model. A designated null state can be exactly δ_0 . This captures the probability model of the current HMM implementation; its automatic grids, moving centers and pruning are estimation procedures.

Each chain follows the existing feature order with unit spacing. There is no cross-modality feature alignment, distance-dependent transition, or learned dependence on `X_f` in this HMM specification. Unrelated sequences require separate chains or an explicitly specified reset.

3.6 Complete joint density and inferential targets

Let $\mathcal{H}$ collect all factor chains and mixture labels, and let $\Theta = (\theta, \psi, \tau)$ collect parameters. Then

$$\begin{aligned} p(Y^A, Y^R, A, R, F^A, F^R, \mathcal{H} \mid \Theta) &= p_A(Y^A \mid A, F^A, \tau^A) p_R(Y^R \mid R, F^R, \tau^R) \\ &\times p(A, R \mid \theta) \prod_{m \in \{A, R\}} \prod_{k=1}^{K_m} p_H(F_k^m, \mathcal{H}_k^m \mid \psi_k^m). \end{aligned} \quad (3.15)$$

An empirical Bayes analysis estimates Θ and targets the joint latent posterior conditional on that estimate. A fully Bayesian analysis additionally specifies $p(\Theta)$ and infers parameter uncertainty. Joint latent inference does not require fully Bayesian treatment of every hyperparameter. Conversely, optimizing all latent variables jointly as point estimates is joint MAP, not posterior uncertainty propagation. With spikes, MAP also needs a precise discrete/continuous parameterization; comparing an atom's mass to a continuous density is not an invariant decision rule.

4 Exact full conditionals

4.1 Gaussian contribution for one loading

Hold other variables and parameters fixed and set

$$e_{it,-k}^m = Y_{it}^m - \sum_{h \neq k} L_{ih}^m F_{th}^m, \quad (4.1)$$

$$a_{ik}^m = \sum_{t:(i,t) \in \Omega_m} \tau_{it}^m (F_{tk}^m)^2, \quad b_{ik}^m = \sum_{t:(i,t) \in \Omega_m} \tau_{it}^m F_{tk}^m e_{it,-k}^m. \quad (4.2)$$

Terms depending on a candidate $v = L_{ik}^m$ reduce exactly to

$$\ell_u(v) = -\frac{1}{2}a_u v^2 + b_u v. \quad (4.3)$$

If $a_u > 0$, define $s_u^2 = 1/a_u$, $x_u = b_u/a_u$. Then $\exp \ell_u(v) \propto \varphi(x_u; v, s_u^2)$. If $a_u = 0$, there is no direct data information and $b_u = 0$; retain the constant likelihood rather than inventing information through an epsilon. Children can still inform the loading in this case.

4.2 Own prior and child contributions

Use $P_c[v]$ for a child's parent vector with the target replaced by v . The exact conditional is

$$p(dv \mid \text{rest}, Y, \Theta) \propto \exp\{\ell_u(v)\} g_u(dv \mid P_u) \prod_{c \in \mathcal{C}_u} g_c(V_c \mid P_c[v]). \quad (4.4)$$

For continuous slabs a zero child uses its spike probability, while a nonzero child uses its slab weight times its slab density. This interpretation also follows by introducing the indicators in (3.9). In particular,

$$p(dA_{ik} \mid \text{rest}, Y, \Theta) \propto e^{\ell_{ik}^A(A_{ik})} g_k^A(dA_{ik} \mid A_{i,<k}) \times \prod_{h>k} g_h^A(A_{ih} \mid A_{i,<h}) \prod_j g_j^R(R_{ij} \mid R_{i,<j}, A_i), \quad (4.5)$$

$$p(dR_{ij} \mid \text{rest}, Y, \Theta) \propto e^{\ell_{ij}^R(R_{ij})} g_j^R(dR_{ij} \mid R_{i,<j}, A_i) \prod_{h>j} g_h^R(R_{ih} \mid R_{i,<h}, A_i). \quad (4.6)$$

Only terms depending on the target need evaluation. The RNA likelihood is constant in (4.5) when R, F^R are fixed; its evidence reaches ATAC through the inferred RNA loadings and the RNA conditional priors. This is posterior feedback under the original ATAC-to-RNA joint prior.

Sampling these full conditionals, together with valid updates for all remaining unknowns, gives a Gibbs or Metropolis-within-Gibbs scheme for one joint target. Replacing samples by conditional means is not Gibbs sampling. Fitting two unrelated reverse-direction conditional priors does not ensure that they are full conditionals of (3.15). Re-estimating parameters during sampling also changes the target; a Bayesian sampler must update those parameters under their conditional distributions, or an EB analysis must separate estimation from inference conditional on the fitted parameters.

5 A joint variational objective and coordinate derivation

5.1 Structured mean-field family

Keep each factor's HMM sequence intact, but initially factorize loading nodes:

$$q = \left\{ \prod_{i,k} q_{ik}^A(A_{ik}) \prod_{i,j} q_{ij}^R(R_{ij}) \right\} \prod_{m,k} q_{F,k}^m(F_k^m, \mathcal{H}_k^m). \quad (5.1)$$

Each loading factor can contain an exact spike and an unrestricted continuous slab. The approximation loses posterior dependence between different loading nodes, but retains their uncertainty and includes every joint-prior term in the objective. The factor sequences are independent across (m, k) under q , not independent across feature positions within a sequence.

For fixed Θ , the evidence lower bound is

$$\mathcal{L}(q, \Theta) = \mathbb{E}_q \log p(Y, A, R, F, \mathcal{H} \mid \Theta) - \mathbb{E}_q \log q. \quad (5.2)$$

It equals $\log p(Y \mid \Theta) - \text{KL}(q \parallel p(\cdot \mid Y, \Theta))$. For this model it can be organized as

$$\begin{aligned} \mathcal{L} = & \sum_m \mathbb{E}_q \log p_m(Y^m \mid L^m, F^m, \tau^m) + \sum_u \mathbb{E}_q \log g_u(V_u \mid P_u) \\ & + \sum_u H(q_u) - \sum_{m,k} \text{KL}(q_{F,k}^m \parallel p_H^m(\cdot \mid \psi_k^m)). \end{aligned} \quad (5.3)$$

Every conditional prior appears *once*. It may enter the updates of several variables because it depends on each of them. This is not double counting in the objective. If parameters have a prior and are point-estimated, add $\log p(\Theta)$ for a MAP parameter objective. A variational distribution over parameters requires its own expected prior and entropy terms instead.

5.2 Derivation, not just the coordinate identity

Holding q_{-u} fixed, collect $f_u(v) = \mathbb{E}_{q_{-u}} \log p(Y, V_u = v, V_{-u}, F, \mathcal{H} \mid \Theta)$. The terms varying with q_u are

$$\int q_u(v) f_u(v) d\nu_u(v) - \int q_u(v) \log q_u(v) d\nu_u(v) = -\text{KL}(q_u \parallel q_u^*) + \log Z_u, \quad (5.4)$$

where $q_u^*(v) = \exp f_u(v) / Z_u$, with a suitable mixed reference measure ν_u . Thus the unrestricted coordinate optimum is

$$\log q_u^*(v) = C + \mathbb{E}_{q_{-u}} \log p(Y, V, F, \mathcal{H} \mid \Theta). \quad (5.5)$$

This standard identity is reviewed by Blei et al. [4]; the remaining expressions below follow by expanding this particular joint model. Exact coordinate maximization cannot decrease (5.2). A Gaussian approximation to a tilted slab, noisy quadrature or incomplete network optimization does not inherit that guarantee automatically.

5.3 Normal-means statistics with posterior uncertainty

Define $m_{ik}^L = \mathbb{E} L_{ik}$, $M_{ik}^L = \mathbb{E} L_{ik}^2$, $m_{tk}^F = \mathbb{E} F_{tk}$, $M_{tk}^F = \mathbb{E} F_{tk}^2$, separately by modality. Using (5.1) and fixed precisions gives

$$a_{ik}^m = \sum_{t:(i,t) \in \Omega_m} \tau_{it}^m M_{tk}^{F,m}, \quad (5.6)$$

$$b_{ik}^m = \sum_{t:(i,t) \in \Omega_m} \tau_{it}^m m_{tk}^{F,m} \left(Y_{it}^m - \sum_{h \neq k} m_{ih}^{L,m} m_{th}^{F,m} \right). \quad (5.7)$$

The coefficient of v^2 uses a *second moment*, not the squared mean. For an independent variational precision factor replace τ by $\mathbb{E}\tau$. The remainder of the coordinate update is

$$\log q_u^*(v) = C - \frac{1}{2}a_u v^2 + b_u v + \mathbb{E} \log g_u(v | P_u) + D_u(v), \quad D_u(v) = \sum_{c \in \mathcal{C}_u} \mathbb{E} \log g_c(V_c | P_c[v]). \quad (5.8)$$

Expectations integrate all other uncertain arguments, never the candidate v . Evaluating a network at posterior means generally differs from the expected log probability in this equation.

The current conditional EBNM approximation effectively uses

$$q_u^{\text{plug}}(dv) \propto \varphi(x_u; v, s_u^2) g_u(dv | \mathbb{E}P_u). \quad (5.9)$$

It both omits D_u and replaces uncertain predictors by their means. An ordinary local EBNM marginal likelihood cannot be substituted for all the joint-prior and entropy terms in (5.3) after this change.

6 A general update for the learned registry

6.1 One normalized equation for any evaluable conditional prior

Assume that the conditional priors admit densities with respect to specified reference measures with fixed supports, and that the following normalizer is positive and finite. These assumptions cover the scalar learned priors in the registry with fixed component supports and positive variances. Define

$$U_u(v) = -\frac{1}{2}a_u v^2 + b_u v + \mathbb{E} \log g_u(v | P_u) + \sum_{c \in \mathcal{C}_u} \mathbb{E} \log g_c(V_c | P_c[v]). \quad (6.1)$$

Then the general mean-field coordinate update and its moments are

$$q_u^*(dv) = \frac{e^{U_u(v)} \nu_u(dv)}{\int e^{U_u(w)} \nu_u(dw)}, \quad \mathbb{E}_{q_u^*} f(V_u) = \frac{\int f(v) e^{U_u(v)} \nu_u(dv)}{\int e^{U_u(v)} \nu_u(dv)}. \quad (6.2)$$

For a spike at zero plus a continuous density, $\nu = \delta_0 + \text{Lebesgue}$. The density at the atom is its probability mass. Its value on the continuous part is the continuous mixture density; one must not add a Gaussian density evaluated at zero to the atom's mass. Other finite atoms are handled by adding their separate mass terms to the integrals. Priors with moving supports or singular parameter-dependent transformations need separate treatment.

Equation (6.2) is a *general update*, but not a universal closed-form solver. It requires conditional log-prior evaluation for arbitrary candidate values and uncertain parent inputs. A function returning only `post_mean`, `post_mean2` and a local marginal likelihood does not expose enough information to evaluate it.

6.2 Reusing an EBNM posterior as a reference distribution

Freeze prior parameters and choose a reference parent value $p_\star$. Let an ordinary normal-means posterior be

$$q_{0u}(dv) \propto \varphi(x_u; v, s_u^2) g_u(dv | p_\star). \quad (6.3)$$

Suppose it is positive everywhere the desired update has positive mass or density. With all densities defined against the same reference measure, put

$$C_u(v) = \mathbb{E} \log g_u(v | P_u) - \log g_u(v | p_\star) + D_u(v). \quad (6.4)$$

Then

$$q_u^*(dv) = \frac{q_{0u}(dv)e^{C_u(v)}}{\mathbb{E}_{q_{0u}} e^{C_u(V_u)}}. \quad (6.5)$$

Thus quadrature or importance sampling from an existing conditional posterior can in principle support every scalar learned family, including EMDN with parent-dependent means and variances. For samples $v_b \sim q_{0u}$, a self-normalized approximation is $\mathbb{E}f(V_u) \approx \sum_b \tilde{w}_b f(v_b)$, $\tilde{w}_b \propto e^{C_u(v_b)}$. This finite-sample ratio is not unbiased, can have highly variable weights, and needs support and tail checks. If spike probability is small, integrating its mass separately is preferable to hoping it appears in a finite sample. The identity is exact; this numerical procedure need not be.

6.3 An exact-target Metropolis-within-Gibbs update

There is an especially simple general route if sampling is acceptable. For an exact scalar full conditional, all other latent variables are fixed. Compute a_u, b_u from their *realized values* as in Section 4, not from the variational moments in (5.6)–(5.7), and take $p_\star = P_u$. Then the correction in (6.5) is just the exact child log likelihood

$$D_u^G(v) = \sum_{c \in \mathcal{C}_u} \log g_c(V_c \mid P_c[v]). \quad (6.6)$$

Propose $v' \sim q_{0u}$, the *ordinary conditional normal-means posterior* under the fixed fitted prior. From the current value v , accept with probability

$$\mathcal{A}(v, v') = \min \left\{ 1, \exp [D_u^G(v') - D_u^G(v)] \right\}. \quad (6.7)$$

To verify it, write the target $t(v) \propto q_{0u}(v)e^{D_u^G(v)}$. The independence Metropolis ratio [3] is $t(v')q_{0u}(v)/[t(v)q_{0u}(v')] = e^{D_u^G(v') - D_u^G(v)}$. The cancellation works for both atom masses and continuous densities on the mixed measure. It does not require differentiation, a Gaussian slab, or a normalizing integral for the tilted target.

For a fixed-parent Gaussian mixture, sampling q_{0u} is straightforward: sample a component using weights proportional to its prior weight times its normal-means evidence, then sample its Gaussian posterior or atom. All scalar learned registry families below have that form once their parameters and parents are fixed. The existing interface would need to expose this posterior, not merely its moments. If a different proposal r is used, the acceptance ratio must include the uncanceled target/proposal terms.

One or several such invariant transitions per coordinate yields Metropolis-within-Gibbs, provided the overall chain has the required support and ergodicity. This is exact-target MCMC, not an exact independent posterior draw in finitely many steps. Strong child constraints may make acceptance poor and mixing slow. For a complete sampler, factor HMM blocks must also be *sampled* using forward-filter/backward-sampling and component/effect draws, with pseudo-data formed from realized current loadings, rather than replaced by posterior means. Parameters must be fixed for conditional EB inference or updated under their Bayesian conditionals. Refitting an EBNM prior on each proposal destroys this cancellation and fixed target unless that parameter update is separately justified.

6.4 Which probability models the registry supplies

The mapping below is based on the current `LearnedPriorType` and solver sources. A Gaussian whose mean is constrained positive still has signed support. A sharpening step during estimation is not itself a different normalized likelihood or a proof of a joint-objective ascent step.

Registry name	Conditional prior	Joint-update consequence
<code>cash</code>	Spike at zero plus zero-centered Gaussian scale mixture; neural mixture weights.	Fixed-component mixture formulas apply; child messages involve the full mixture density or explicit component labels.
<code>lcash</code>	Same scale mixture; linear softmax weights.	Same generic mixture update, with a simpler gate.
<code>po_lcash</code>	Same scale mixture; ordered-logistic probabilities.	Same generic mixture update; respect ordered cut-point constraints.
<code>cgb</code>	Spike plus one signed Gaussian; neural slab probability, global mean and variance.	Section 8 gives its simpler scalar spike/tilted-Gaussian update.
<code>cgb_sharp</code>	Same probability family as CGB at fixed parameters; variance sharpening during fitting.	Conditional inference has the CGB form; the sharpening rule needs a separate objective justification.
<code>cgb_sharp_2</code>	Spike plus two signed Gaussians with global positive/negative mean constraints and neural weights.	General mixture form; sign-constrained means do not truncate the slabs.
<code>emdn</code>	Gaussian mixture whose weights, means and variances depend on covariates. No exact spike.	Include parent effects on every mixture parameter in child messages.
<code>spiked_emdn</code>	Exact zero spike plus EMDN Gaussian slabs.	Same EMDN message with a separate spike mass.
<code>hmm, hmm_pos, hmm_neg</code>	Sequence priors with latent Markov states; ignore covariates.	Treat the dependent sequence as a block. They are not scalar covariate-density priors merely because they appear in the learned registry.

For scalar priors, automatic component grids should be fixed while deriving one target, or changed only through evaluated model proposals. Conditional variances must remain positive for continuous components. Fixed-component weights can change with parents without moving the underlying atom support.

6.5 A common augmented-mixture variational update

All scalar entries in the table can be represented, at the probability-model level, using a component label B_u :

$$p(B_u = h, V_u \mid P_u) = w_{uh}(P_u) G_{uh}(V_u \mid P_u), \quad \sum_h w_{uh}(P_u) = 1. \quad (6.8)$$

The labels in this subsection are distinct from the HMM mixture labels. For continuous components, write $G_{uh} = \mathcal{N}(\mu_{uh}(P_u), d_{uh}^2(P_u))$; an optional component is the fixed zero atom. Use the variational family $\prod_u q_u(V_u, B_u) \prod_{m,k} q_{F,k}^m$, so that the value and its own label are *not* forced independent. Parents enter the model through their values, not through these auxiliary labels.

This is a different variational family on an augmented joint model. In general,

$$\mathbb{E} \log \sum_h w_h G_h \neq \log \sum_h \exp\{\mathbb{E} \log(w_h G_h)\}.$$

The augmented update must not be advertised as the exact collapsed coordinate (6.2) when parents are uncertain. Both target the same generative marginal model, but their variational approximations differ.

Let $r_{ch} = q(B_c = h)$, and for a continuous child component let $m_{ch} = \mathbb{E}(V_c \mid B_c = h)$, $M_{ch} = \mathbb{E}(V_c^2 \mid B_c = h)$. The downstream contribution in the augmented family is

$$D_u^{\text{aug}}(v) = \sum_{c \in \mathcal{C}_u} \sum_h r_{ch} \mathbb{E}_{P_c \setminus V_u} [\log w_{ch}(P_c[v]) + J_{ch}(P_c[v])], \quad (6.9)$$

where, for a Gaussian child component,

$$J_{ch}(p) = -\frac{1}{2} \log(2\pi d_{ch}^2(p)) - \frac{M_{ch} - 2\mu_{ch}(p)m_{ch} + \mu_{ch}^2(p)}{2d_{ch}^2(p)}. \quad (6.10)$$

For the fixed spike component, $J_{c0} = 0$ on its supported event. This uses the per-node mean-field independence from parents while preserving $q(V_c, B_c)$. It handles parent-dependent EMDN means and variances, not only weight-dependent CGB gates.

For each continuous component of the target, define

$$A_h = a_u + \mathbb{E}[d_{uh}^{-2}(P_u)], \quad (6.11)$$

$$B_h = b_u + \mathbb{E}[\mu_{uh}(P_u)d_{uh}^{-2}(P_u)], \quad (6.12)$$

$$C_h = \mathbb{E} \log w_{uh}(P_u) - \frac{1}{2} \mathbb{E} \log(2\pi d_{uh}^2(P_u)) - \frac{1}{2} \mathbb{E}[\mu_{uh}^2(P_u)d_{uh}^{-2}(P_u)]. \quad (6.13)$$

Assume these expectations are finite and $A_h > 0$. Put $v_h = 1/A_h$, $m_h = B_h/A_h$, $I_{rh} = \int v^r \varphi(v; m_h, v_h) e^{D_u^{\text{aug}}(v)} dv$. Completing the square gives unnormalized component masses

$$W_h = \exp \left\{ C_h + \frac{B_h^2}{2A_h} \right\} \sqrt{\frac{2\pi}{A_h}} I_{0h}, \quad h \text{ continuous}, \quad (6.14)$$

$$W_0 = \exp\{\mathbb{E} \log w_{u0}(P_u) + D_u^{\text{aug}}(0)\} \quad \text{if a zero atom is present.} \quad (6.15)$$

There is no normal-means evidence constant in (6.15) because the common likelihood factor $\exp \ell_u(v)$, rather than a normalized Gaussian in x , was used for *all* components. After setting $r_{uh} = W_h / \sum_b W_b$,

$$q_u^*(dv, B_u = h) = r_{uh} \frac{\varphi(v; m_h, v_h) e^{D_u^{\text{aug}}(v)}}{I_{0h}} dv \quad (h \text{ continuous}), \quad (6.16)$$

with mass r_{u0} at the spike. Therefore $\mathbb{E}V_u = \sum_{h \text{ cont}} r_{uh} I_{1h}/I_{0h}$ and $\mathbb{E}V_u^2 = \sum_{h \text{ cont}} r_{uh} I_{2h}/I_{0h}$. This is a common tilted-Gaussian-mixture update for every scalar learned registry family, including EMDN. It remains nonconjugate through the downstream tilt and uncertain-parent expectations, but no new family-specific derivation is needed for each choice of gate or covariate-dependent Gaussian parameters.

For fixed Gaussian components, simplify the expectations in A_h, B_h, C_h to their constant values and expected log weights. For CGB, the zero and Gaussian components have disjoint supports up to a measure-zero event, so these equations reduce to Section 8. For overlapping Gaussian mixtures, component augmentation genuinely changes the chosen mean-field approximation.

The augmented parameter update is likewise general:

$$\theta_u^{\text{new}} \in \arg \max_{\theta_u} \sum_i \mathbb{E}_{q(P_{iu})q(V_{iu}, B_{iu})} [\log w_{uB_{iu}}(P_{iu}; \theta_u) + \log G_{uB_{iu}}(V_{iu} \mid P_{iu}; \theta_u)] + \log p(\theta_u). \quad (6.17)$$

For EMDN this trains weights, means and variances together using latent posterior expectations; for fixed-component mixtures only the weight-network terms vary. It is not generally the old isolated EBNM marginal-likelihood training call. The augmented ELBO must retain component-label entropy and expected complete-data prior terms consistently.

6.6 Sequence priors and a general factor-graph statement

For a block V_B , the same principle is

$$\log q_B^*(v_B) = C + \mathbb{E}_{q_{-B}} \sum_{f: \text{scope}(f) \cap B \neq \emptyset} \log f(v). \quad (6.18)$$

Sum every joint factor touching the block, not just scalar parent/child terms. This covers HMM priors without falsely making their coordinates independent. In the requested model the HMM is on F , where it has no loading children, so Section 9 gives the direct block update. If an HMM were placed on L , it would couple ordered cells; those states and neighbors must be included. When its downstream correction factorizes over positions, smoothing can still use emissions obtained by one-dimensional component integrals with the extra tilt. More general cross-position corrections need a different block solver. An HMM posterior proposal can also be Metropolis-corrected using the full block downstream log-likelihood difference, provided the proposal can be sampled and the required support condition holds.

6.7 The reusable interface implied by the mathematics

A general solver needs, conceptually:

1. Frozen prior parameters, normalized conditional log densities and explicit atom masses at arbitrary parent values, with fixed preprocessing.
2. A representation of the prior and posterior components, and either posterior sampling or numerical integration support.
3. Expected log-prior and parameter-gradient evaluations under uncertain parents and child values, if using variational inference.
4. A joint dependency graph and one consistent objective, including any component-label entropy, HMM block terms and declared penalties.

The experimental implementation described in Section C now supplies the density, component-sampling and joint-graph interfaces for the scalar families; the variational extensions remain specifications. The current registry can select the component family; its moment-returning `fit` contract alone is insufficient for a universal joint update. The same scalar formulas apply to a feature-factor entry if a learned conditional prior is placed on F , after substituting its factor-side Gaussian coefficients and its actual parent/child graph. With fixed external feature covariates and no factor children, the update reduces to the ordinary conditional EBNM problem. Latent self-covariates on F introduce the same expected-parent and child-message requirements as on L .

7 Gaussian, exponential and atomic conditional mixtures

This section specializes the joint update to the distribution families of primary interest. The same construction covers a Gaussian mixture, an exponential mixture, either family with point masses, and a mixture containing both continuous families. The normalized normal-means proposal is available analytically; its hierarchical correction need not be conjugate.

7.1 A normalized family with fixed supports

For node u in cell i , let $h_{iu} = (P_{iu}, X_i)$ collect its permitted latent parents and fixed observed covariates. Suppress these indices where unambiguous. Introduce a component label B with law

$$g(dv \mid h) = \sum_{j \in \mathcal{A}} w_j(h) \delta_{c_j}(dv) + \sum_{j \in \mathcal{G}} w_j(h) \varphi(v; \mu_j(h), d_j^2(h)) dv + \sum_{j \in \mathcal{E}} w_j(h) \lambda_j(h) e^{-\lambda_j(h)v} \mathbf{1}\{v > 0\} dv. \quad (7.1)$$

Here the three label sets are disjoint, $w_j \geq 0$, $\sum_j w_j = 1$, $d_j^2 > 0$, and $\lambda_j > 0$. Any set may be empty, provided at least one component remains. A zero spike corresponds to a fixed $c_j = 0$. The atoms and component supports do not depend on parents or trainable parameters in this derivation. Moving an atom with a parent would introduce deterministic support constraints and invalidate the simple density cancellations below without further analysis.

The augmented conditional is $g(dv, j \mid h) = w_j(h) G_j(dv \mid h)$. Use Lebesgue measure on each continuous component and unit point mass on each atomic component as the reference measure. Equivalently, a collapsed prior uses Lebesgue measure plus the distinct fixed atoms. At an atom its Radon–Nikodym derivative is the total atomic probability; continuous densities evaluated at that point are not added to the atomic probability. This convention removes any need to approximate a spike by a narrow Gaussian.

Neural networks may parameterize weights by softmax and positive scales or rates by suitable positive transformations. A graph neural network is also admissible if its latent inputs respect the specified directed acyclic graph. If it introduces additional cross-cell dependencies, the child sets and parallel update schedule must change accordingly. Fixed side information may enter every conditional without being sampled. In all cases, each head must specify a normalized probability distribution.

7.2 Analytic normal-means proposals and moments

Condition on current parents, other latent variables and parameters, and write the observation contribution as $\ell(v) = -av^2/2 + bv$, with $a \geq 0$ and $a = 0 \Rightarrow b = 0$. For every component define

$$I_j = \int e^{\ell(v)} G_j(dv), \quad Q_j(dv) = I_j^{-1} e^{\ell(v)} G_j(dv), \quad \tilde{w}_j = \frac{w_j I_j}{\sum_r w_r I_r}. \quad (7.2)$$

The exact own-prior normal-means posterior is $q_0(dv, j) = \tilde{w}_j Q_j(dv)$. The same unnormalized likelihood $e^{\ell(v)}$ is used for every label; one must not combine these I_j with normalized observation densities for only some components.

Atomic components. For $G_j = \delta_{c_j}$,

$$\log I_j = -ac_j^2/2 + bc_j, \quad Q_j = \delta_{c_j}, \quad \mathbb{E}_{Q_j} V = c_j, \quad \mathbb{E}_{Q_j} V^2 = c_j^2. \quad (7.3)$$

In particular $I_j = 1$ for the zero spike. Its posterior probability still changes through normalization against the continuous components.

Gaussian components. Let $G_j = \mathcal{N}(\mu_j, d_j^2)$ and $t_j = 1 + ad_j^2$. Completing the square gives

$$V_j = \frac{d_j^2}{t_j}, \quad M_j = \frac{\mu_j + bd_j^2}{t_j}, \quad Q_j = \mathcal{N}(M_j, V_j), \quad (7.4)$$

$$\log I_j = -\frac{1}{2} \log t_j + \frac{-a\mu_j^2/2 + b\mu_j + b^2d_j^2/2}{t_j}. \quad (7.5)$$

Thus $\mathbb{E}_{Q_j} V = M_j$ and $\mathbb{E}_{Q_j} V^2 = M_j^2 + V_j$. The expression for $\log I_j$ avoids subtracting large μ_j^2/d_j^2 terms when the slab is narrow.

Exponential components. Let $G_j(dv) = \lambda_j e^{-\lambda_j v} \mathbf{1}\{v > 0\} dv$. For $a > 0$, put $z_j = (b - \lambda_j)/\sqrt{a}$ and $m_j = (b - \lambda_j)/a$. Since

$$-av^2/2 + (b - \lambda_j)v = -a(v - m_j)^2/2 + (b - \lambda_j)^2/(2a),$$

integration over the positive half-line gives

$$Q_j = \mathcal{N}_{[0, \infty)}(m_j, a^{-1}), \quad (7.6)$$

$$\log I_j = \log \lambda_j + \frac{1}{2} \log(2\pi/a) + \frac{1}{2} z_j^2 + \log \Phi(z_j). \quad (7.7)$$

Writing $r(z) = \varphi(z)/\Phi(z)$, its first two moments are

$$\mathbb{E}_{Q_j} V = \frac{z_j + r(z_j)}{\sqrt{a}}, \quad \mathbb{E}_{Q_j} V^2 = \frac{1 + z_j^2 + z_j r(z_j)}{a}. \quad (7.8)$$

These follow either from truncated-normal moments or by differentiating I_j with respect to b . If $a = b = 0$, evaluate the limit directly:

$$I_j = 1, \quad Q_j = \text{Exp}(\lambda_j), \quad \mathbb{E}V = \lambda_j^{-1}, \quad \mathbb{E}V^2 = 2\lambda_j^{-2}.$$

This case occurs for a loading without a direct modality likelihood. Its children can nevertheless inform its posterior.

For negative z , an equivalent expression avoids cancellation in the log normalizer:

$$\log I_j = \log \lambda_j + \frac{1}{2} \log\{\pi/(2a)\} + \log \text{erfcx}(-z_j/\sqrt{2}), \quad (7.9)$$

where $\text{erfcx}(x) = e^{x^2} \text{erfc}(x)$. Use log-CDF functions for the other branch and log-sum-exp for mixture weights. The moment identities in (7.8) can still suffer cancellation for large negative z ; stable tail expansions or specialized truncated-normal routines are needed in numerical solvers. For example, if $x = -z > 0$, the unnormalized moments of $T = \sqrt{a}V$ are

$$K_n(x) = \int_0^\infty t^n e^{-xt - t^2/2} dt \sim x^{-n-1} \sum_{k=0}^\infty \frac{(-1)^k (n+2k)!}{2^k k! x^{2k}} \quad (x \rightarrow \infty).$$

Use $K_1/(\sqrt{a}K_0)$ and $K_2/(aK_0)$ for the first two moments, truncating this asymptotic expansion in the far tail. The expansion follows by setting $y = xt$, expanding $e^{-y^2/(2x^2)}$, and integrating each term against e^{-y} . It is an asymptotic expansion, not a convergent series at every x . There is no justification for replacing $a = 0$ by an arbitrary positive precision.

A reflected exponential with fixed orientation $\kappa_j \in \{-1, 1\}$ has density $\lambda_j e^{-\lambda_j \kappa_j v} \mathbf{1}\{\kappa_j v > 0\}$. Apply (7.6)–(7.8) to $T = \kappa_j V$ with b replaced by $\kappa_j b$. Multiply the first moment by κ_j ; leave the second moment unchanged. The sign of a Gaussian mean, by contrast, does not restrict its support.

Mixture summaries. Without children, the first two posterior moments are

$$\mathbb{E}_{q_0} V = \sum_j \tilde{w}_j \mathbb{E}_{Q_j} V, \quad \mathbb{E}_{q_0} V^2 = \sum_j \tilde{w}_j \mathbb{E}_{Q_j} V^2.$$

With a zero spike and otherwise continuous components, $\Pr(V = 0) = \tilde{w}_0$. Under the convention $\text{lfsr} = \min\{\Pr(V \leq 0), \Pr(V \geq 0)\}$, the spike contributes to both probabilities; a positive exponential contributes only to the second, and a Gaussian contributes its appropriate normal-CDF probability. These are marginal posterior summaries, not classification based on the largest component density.

7.3 Exact hierarchical correction

During an augmented coordinate move keep each child's value V_c and label B_c fixed. Define

$$D_u(v) = \sum_{c \in \text{ch}(u)} \log\{w_{cB_c}(h_c[v])G_{cB_c}(V_c | h_c[v])\}. \quad (7.10)$$

The product is evaluated on the reference measures specified above. For each component put

$$J_{jr} = \int v^r e^{D_u(v)} Q_j(dv), \quad r = 0, 1, 2.$$

Assuming a finite positive normalizer, the exact full conditional is

$$p(dv, j | \text{rest}) = \frac{w_j I_j}{\sum_h w_h I_h J_{h0}} e^{D_u(v)} Q_j(dv). \quad (7.11)$$

Its label probabilities are proportional to $w_j I_j J_{j0}$, and its first two moments are obtained by replacing J_{j0} in the numerator with J_{j1} and J_{j2} , respectively. For an atom, $J_{jr} = c_j^r e^{D_u(c_j)}$, with $c_j^0 = 1$. This supplies a one-dimensional quadrature route with exact atomic masses. The tilted continuous components generally cease to be Gaussian or truncated Gaussian.

Alternatively draw $(v', j') \sim q_0$ and accept with probability

$$\alpha = \min\{1, \exp[D_u(v') - D_u(v)]\}. \quad (7.12)$$

To verify the rule, write the target as $\pi(v, j) = q_0(v, j) e^{D_u(v)} / Z_D$. The forward accepted probability density is

$$\pi(v, j) q_0(v', j') \alpha = \frac{q_0(v, j) q_0(v', j')}{Z_D} \min\{e^{D_u(v)}, e^{D_u(v')}\},$$

which is symmetric in the two states. Rejections complete the transition kernel, establishing detailed balance. Fixed-support mixture augmentation therefore permits exact transitions between spikes and slabs without a reversible-jump construction. This is the standard Metropolis principle [3], specialized to analytic normal-means proposals. It establishes invariance; irreducibility, adequate mixing and Monte Carlo precision must be assessed separately.

For a Gaussian child component, include

$$\log w_{cj}(h) - \frac{1}{2} \log(2\pi d_{cj}^2(h)) - \frac{(V_c - \mu_{cj}(h))^2}{2d_{cj}^2(h)}.$$

For a positive exponential child component, include

$$\log w_{cj}(h) + \log \lambda_{cj}(h) - \lambda_{cj}(h) V_c \quad (V_c > 0). \quad (7.13)$$

For a fixed atomic child on its support, only the log weight remains. The exponential rate normalizer and the Gaussian variance normalizer are essential whenever they depend on the updated parent. In particular, weights alone are insufficient for EMDN.

The obstruction to conjugacy is already present in the single-Gaussian example $V_c \mid v \sim \mathcal{N}(v^2, s^2)$, whose child log density contains $-(V_c - v^2)^2/(2s^2)$. Removing a spike does not remove this nonlinear term. Conversely, linear means and constant variances in a Gaussian loading DAG preserve Gaussian scalar conditionals, conditional on the feature factors.

7.4 Specialization to CGB, sharp CGB and two-slab CGB

For the package’s CGB prior,

$$g(dv \mid h) = \{1 - \pi(h)\}\delta_0(dv) + \pi(h)\mathcal{N}(dv; \mu, d^2).$$

The normal-means masses are $1 - \pi$ and πI_G , with I_G from (7.5). The hierarchical slab log odds are

$$\log \frac{\Pr(B = 1 \mid \text{rest})}{\Pr(B = 0 \mid \text{rest})} = \text{logit } \pi + \log I_G + \log J_{G0} - D_u(0). \quad (7.14)$$

Its conditional slab moments are J_{G1}/J_{G0} and J_{G2}/J_{G0} . If a CGB child has gate $\pi_c(v) = \text{sigmoid } \eta_c(v)$, its variable log contribution is $B_c \eta_c(v) - \log\{1 + \exp \eta_c(v)\}$. The same expression applies to its sharp variant at fixed slab parameters.

Sharp CGB uses an effective slab variance $d^2 = \omega s^2$, with $\omega > 0$, in the preceding formulas. A small positive variance is a continuous slab, not a second point mass. If the base variance is freely estimated, a fixed ω merely reparameterizes the same family. Persistent sharpening requires a fixed scale, an explicit constraint or a specified regularizer. Repeatedly multiplying an unconstrained variance estimate by ω is not the unpenalized joint M-step.

The two-slab version has

$$g(dv \mid h) = \pi_0(h)\delta_0(dv) + \pi_+(h)\mathcal{N}(dv; \mu_+, d_+^2) + \pi_-(h)\mathcal{N}(dv; \mu_-, d_-^2), \quad (7.15)$$

where $\mu_+ > 0$, $\mu_- < 0$, and the weights sum to one. The means are global parameters, and both Gaussians have support on the whole real line. Its three posterior masses are proportional to

$$\pi_0 e^{D_u(0)}, \quad \pi_+ I_+ J_{+0}, \quad \pi_- I_- J_{-0}.$$

The moments follow by the same weighted integrals. With $d_\pm^2 = \omega s_\pm^2$, this is the fixed-parameter model of `cgb_sharp_2`. Setting $\pi_0 = 0$ gives a two-Gaussian variant without a spike; this is a distributional specialization, not an additional registry name currently exposed by the package.

Setting the spike weight to zero in a one-slab CGB leaves a single global Gaussian. Its gate disappears, so that node’s own conditional prior no longer depends on parents. Retaining parent dependence without a spike requires multiple changing weights or parent-dependent means or variances, as in EMDN. CASH and LCASH are further Gaussian-mixture specializations with fixed zero-centered scale grids and their respective gate functions.

7.5 Variational updates for the extended family

For an alternative approximation, use the augmented mean-field family

$$q = \prod_u q_u(V_u, B_u) \prod_{m,k} q_{F,k}^m$$

from Section 6.5. Here a, b are the expected-likelihood coefficients from Section 5, rather than realized-value Gibbs coefficients. Define $D_u^q(v)$ as the expected log child contribution under all other variational factors. Then

$$q_u^*(dv, j) \propto \exp\{\ell(v) + \mathbb{E}_q \log[w_j(h)G_j(v | h)] + D_u^q(v)\} \nu_j(dv). \quad (7.16)$$

All following expectations require finiteness. For a Gaussian component this is the tilted Gaussian construction in (6.14)–(6.16), with expected inverse variances and expected precision-weighted means. Replacing these by the parameters evaluated at mean parents is generally incorrect.

For an exponential component set

$$B_j = b - \mathbb{E}_q \lambda_j(h), \quad C_j = \mathbb{E}_q \log w_j(h) + \mathbb{E}_q \log \lambda_j(h). \quad (7.17)$$

Its unnormalized log coordinate density on $v > 0$ is exactly $-av^2/2 + B_j v + C_j + D_u^q(v)$. For $a > 0$, define

$$H_j = \sqrt{2\pi/a} \exp\{B_j^2/(2a)\} \Phi(B_j/\sqrt{a}), \quad (7.18)$$

$$Q_j^q = \mathcal{N}_{[0,\infty)}(B_j/a, a^{-1}), \quad J_{jr}^q = \mathbb{E}_{Q_j^q}\{V^r e^{D_u^q(V)}\}. \quad (7.19)$$

The component mass is $W_j = e^{C_j} H_j J_{j0}^q$; conditional moments are J_{j1}^q/J_{j0}^q and J_{j2}^q/J_{j0}^q . When $a = b = 0$, let $\bar{\lambda}_j = \mathbb{E}_q \lambda_j(h) > 0$. Then $H_j = 1/\bar{\lambda}_j$ and $Q_j^q = \text{Exp}(\bar{\lambda}_j)$, with the same tilted integral formulas. Notice that e^{C_j} includes $\exp(\mathbb{E} \log \lambda_j)$; it cannot be replaced by $\mathbb{E} \lambda_j$ in the component mass.

For an atomic component at c_j , the unnormalized mass is

$$W_j = \exp\{\ell(c_j) + \mathbb{E}_q \log w_j(h) + D_u^q(c_j)\}.$$

Normalize all Gaussian, exponential and atomic masses together. For an exponential child component with $r_{cj} = q(B_c = j)$ and $m_{cj} = \mathbb{E}_q(V_c | B_c = j)$, its contribution to $D_u^q(v)$ is

$$r_{cj} \mathbb{E}_{P_c \setminus V_u} \{\log w_{cj}(h_c[v]) + \log \lambda_{cj}(h_c[v]) - \lambda_{cj}(h_c[v]) m_{cj}\}. \quad (7.20)$$

This factorization uses the stated mean-field independence between a child node and its parents. A row-structured approximation instead retains their joint expectation.

These are exact coordinate equations within the stated variational family, provided the expectations and integrals are computed exactly. Monte Carlo expectations, quadrature and parametric approximations to tilted components are separate numerical approximations. The augmented and collapsed mean-field problems are not interchangeable for overlapping mixtures. An implemented variational algorithm must evaluate the corresponding entropy and expected joint terms consistently.

7.6 Learning the conditional distributions

Let $\mathcal{Q}_{\text{old}}$ be either the joint latent posterior under old parameters or a specified variational approximation. A node's complete-data objective is

$$\mathcal{M}_u(\theta_u) = \sum_i \mathbb{E}_{\mathcal{Q}_{\text{old}}} \{\log w_{uB_{iu}}(h_{iu}; \theta_u) + \log G_{uB_{iu}}(V_{iu} | h_{iu}; \theta_u)\} + \mathcal{R}_u(\theta_u). \quad (7.21)$$

Here $\mathcal{R}_u$ is an explicitly specified regularizer or log parameter prior; it is zero for the unpenalized empirical-Bayes objective. The expectation is held fixed while optimizing θ_u . Posterior samples

must retain parents, values and labels from the same joint draw. Latent posterior values are not new noisy observations for another EBNM deconvolution step.

For global component parameters, write $n_j = \sum_i \mathbb{E} \mathbf{1}\{B_{iu} = j\}$, $s_j = \sum_i \mathbb{E}[V_{iu} \mathbf{1}\{B_{iu} = j\}]$, and $t_j = \sum_i \mathbb{E}[V_{iu}^2 \mathbf{1}\{B_{iu} = j\}]$. Without regularization and with positive occupancy, Gaussian components have the unconstrained updates

$$\mu_j^{\text{new}} = s_j/n_j, \quad (d_j^2)^{\text{new}} = t_j/n_j - (s_j/n_j)^2. \quad (7.22)$$

If the mean is constrained to a closed interval $\mathcal{I}_j$, project s_j/n_j onto that interval and use $t_j/n_j - 2\mu_j^{\text{new}} s_j/n_j + (\mu_j^{\text{new}})^2$ for the variance. A strict sign constraint can have only a boundary supremum; using $[\epsilon, \infty)$ or $(-\infty, -\epsilon]$ with fixed $\epsilon > 0$ specifies an attained constrained problem. Variance bounds are enforced by projection onto the specified positive interval. For a global exponential rate,

$$\lambda_j^{\text{new}} = n_j/s_j \quad (7.23)$$

when $s_j > 0$; bounds or regularization are required at degeneracies. Unoccupied components do not have identified parameter updates. For global weights, $w_j^{\text{new}} = n_j/N$.

Neural conditional parameters generally require numerical optimization of (7.21). For a Gaussian draw their gradients with respect to μ and $\log d$ are

$$\frac{v - \mu}{d^2}, \quad \frac{(v - \mu)^2}{d^2} - 1;$$

for an exponential draw the gradient with respect to $\log \lambda$ is $1 - \lambda v$. A softmax logit receives $\mathbf{1}\{B = j\} - w_j(h)$. The network chain rule then accounts for its chosen parameterization, including shared hidden layers. These gradients describe complete-data learning, not the derivative of an isolated normal-means marginal likelihood.

Exact EM increases the observed-data likelihood when its E-step is exact and its M-step increases the corresponding objective. Finite Monte Carlo EM need not do so, even if it improves the fixed-sample objective. Variational coordinate ascent instead concerns its particular ELBO. Mixture variance collapse, rate divergence and overfitting by flexible heads motivate explicit admissible parameter sets and held-out tuning; they are not resolved by the algebra of a scalar proposal.

7.7 Scope of the derivation and implementation

The formulas above supply exact normal-means proposals and invariant hierarchical coordinate kernels for the stated families. They do not claim an analytic general full conditional or uniformly fast mixing. An exponential mixture on the positive half-line has a decreasing continuous density:

$$\frac{d}{dv} \sum_j w_j \lambda_j e^{-\lambda_j v} = - \sum_j w_j \lambda_j^2 e^{-\lambda_j v} < 0 \quad (v > 0).$$

Consequently, it is not a universal family for positive conditional densities and cannot represent a continuous mode away from zero. Adding a zero spike does not change this restriction. The choice between Gaussian and exponential components therefore changes both support and shape assumptions, rather than only computational convenience. Their HMM counterpart remains a sequence block update (Section 9); the loading hierarchy does not introduce children of F when feature factors are not inputs to the loading priors.

The experimental code currently implements Gaussian and fixed-atom conditional mixtures across the eight scalar learned registry families. Exponential-mixture inference and its variational counterpart in this section are derived extensions, not newly implemented solver options. The independent verification script described in Section F tests their local identities and the TwinEB factorization below.

8 Normalized CGB updates

8.1 Simplifying the downstream term

Let $\gamma_c = q(Z_c = 1)$ denote a child's posterior slab probability. Because CGB's μ_c, d_c^2 are global and independent of parents, its slab-density term is constant in the parent's candidate v . Under (5.1), we may therefore use the following representative of D_u , dropping additive constants:

$$D_u(v) = \sum_{c \in \mathcal{C}_u} [(1 - \gamma_c) \mathbb{E} \log(1 - \pi_c(P_c[v])) + \gamma_c \mathbb{E} \log \pi_c(P_c[v])]. \quad (8.1)$$

The expectation is only over the child's other parents. This formula uses independence between a child's loading and its parents under the chosen variational family. It would require modification for a structured joint loading approximation. A parent-dependent slab mean or variance also adds expected child slab-density terms; they cannot then be dropped.

For an exact scalar Gibbs conditional, use actual parents, no expectations, and $\gamma_c = Z_c$ in (8.1). Conditional on the other loadings the same normalized formulas below apply. For a continuous slab, $Z_c = \mathbf{1}\{V_c \neq 0\}$ almost surely.

Define the target's expected log gate weights

$$\eta_{u0} = \mathbb{E} \log(1 - \pi_u(P_u)), \quad \eta_{u1} = \mathbb{E} \log \pi_u(P_u). \quad (8.2)$$

Their exponentials need not sum to one. Replacing them by $1 - \pi_u(\mathbb{E}P_u)$ and $\pi_u(\mathbb{E}P_u)$ is an additional approximation.

8.2 Spike and tilted Gaussian slab

Suppress the node index and assume $a > 0$. Gaussian multiplication gives

$$v_0 = (s^{-2} + d^{-2})^{-1}, \quad m_0 = v_0(x/s^2 + \mu/d^2), \quad (8.3)$$

$$\varphi(x; v, s^2) \varphi(v; \mu, d^2) = \varphi(x; \mu, s^2 + d^2) \varphi(v; m_0, v_0). \quad (8.4)$$

Let

$$I_r = \int_{\mathbb{R}} v^r \varphi(v; m_0, v_0) \exp\{D(v)\} dv, \quad r = 0, 1, 2. \quad (8.5)$$

The unnormalized posterior masses and their normalization are

$$w_0 = e^{\eta_0} \varphi(x; 0, s^2) e^{D(0)}, \quad (8.6)$$

$$w_1 = e^{\eta_1} \varphi(x; \mu, s^2 + d^2) I_0, \quad \gamma = \frac{w_1}{w_0 + w_1}. \quad (8.7)$$

Consequently the exact optimum within the unrestricted scalar family is

$$q^*(dv) = (1 - \gamma) \delta_0(dv) + \gamma \frac{\varphi(v; m_0, v_0) e^{D(v)}}{I_0} dv. \quad (8.8)$$

Its moments are

$$\mathbb{E}V = \gamma I_1 / I_0, \quad \mathbb{E}V^2 = \gamma I_2 / I_0, \quad \text{Var}(V) = \gamma I_2 / I_0 - (\gamma I_1 / I_0)^2. \quad (8.9)$$

The posterior null probability is $1 - \gamma$. These moments alone do not specify a general non-Gaussian slab. To evaluate future expected log gates, the algorithm must retain a representation

of that slab (for example, quadrature nodes and weights, or samples), or declare an additional approximation.

An equivalent log-odds form isolates the downstream correction:

$$\text{logit } \gamma = \eta_1 - \eta_0 + \log \frac{\varphi(x; \mu, s^2 + d^2)}{\varphi(x; 0, s^2)} + \log I_0 - D(0). \quad (8.10)$$

With no children, choose $D = 0$: $I_0 = 1, I_1 = m_0, I_2 = m_0^2 + v_0$, recovering the usual spike-and-Gaussian posterior, with expected-log gate weights for uncertain parents. With fixed parents this is ordinary CGB EBNM. The last RNA component has no children, but its own parent uncertainty remains.

If $a = 0$, replace the base Gaussian by $\mathcal{N}(\mu, d^2)$ directly and use $w_0 = e^{\eta_0 + D(0)}$, $w_1 = e^{\eta_1} I_0$. This is also the appropriate limit of the likelihood-kernel derivation. Compute weights in log space.

8.3 Positive and other slabs

For a positive Gaussian slab, the ordinary un-tilted slab posterior is $\mathcal{N}_+(m_0, v_0)$. Its marginal normal-means likelihood is

$$B_+(x) = \varphi(x; \mu, s^2 + d^2) \frac{\Phi(m_0/\sqrt{v_0})}{\Phi(\mu/d)}. \quad (8.11)$$

Use $I_r^+ = \mathbb{E}_{V \sim \mathcal{N}_+(m_0, v_0)}[V^r e^{D(V)}]$, and replace $\varphi(x; \mu, s^2 + d^2) I_0$ in (8.7) by $B_+(x) I_0^+$. The moment formula is otherwise unchanged. This is a possible positive *loading* model; selecting `hmm_pos` for F does not make the existing CGB loading slab positive.

For a parent-independent exponential slab with rate $\lambda > 0$, its un-tilted posterior is $\mathcal{N}_+(x - \lambda s^2, s^2)$ and its evidence is

$$B_{\text{exp}}(x) = \lambda \exp\{-\lambda x + \lambda^2 s^2/2\} \Phi(x/s - \lambda s). \quad (8.12)$$

The same spike/slab evidence-times-tilt construction applies. For general parent-dependent slabs, start from (5.8); neither this Gaussian base nor the simplified CGB message is generally available.

8.4 Why the generic update is difficult

For a scalar full conditional, $D(v)$ can be evaluated with all other variables fixed, so normalization is one-dimensional, even for a neural gate. For a variational update, evaluating $D(v)$ and η_z also requires expectations over uncertain parent vectors. There is usually no analytic formula for these logistic/neural expectations. Deterministic quadrature over all parents scales poorly; Monte Carlo replaces it with sampling error.

A possible numerical calculation draws parents from their current variational factors, reuses those draws across candidate v , averages $\log$ gate probabilities, and then integrates (8.5) in one dimension. It must preserve the exact spike weight and check normalization and tails. Exponentiating an unbiased estimate of $\mathbb{E} \log g$ is not an unbiased estimate of its exponential. Finite Monte Carlo coordinate steps are therefore approximations to CAVI, not exact monotone updates. Arbitrary neural gates can also produce multiple slab modes.

9 Factor HMM updates remain tractable

9.1 Normal-means reduction for an entire factor

Because loading priors do not depend on F , no loading-child term enters a factor update. Under (5.1), define

$$a_{tk}^{F,m} = \sum_{i:(i,t) \in \Omega_m} \tau_{it}^m M_{ik}^{L,m}, \quad (9.1)$$

$$b_{tk}^{F,m} = \sum_{i:(i,t) \in \Omega_m} \tau_{it}^m m_{ik}^{L,m} \left(Y_{it}^m - \sum_{h \neq k} m_{ih}^{L,m} m_{th}^{F,m} \right), \quad (9.2)$$

$$(s_{tk}^{F,m})^2 = (a_{tk}^{F,m})^{-1}, \quad x_{tk}^{F,m} = b_{tk}^{F,m} / a_{tk}^{F,m}. \quad (9.3)$$

For positive $a_{tk}^{F,m}$,

$$q_{F,k}^{m*}(dF, Q, C) \propto p_H(dF, Q, C \mid \psi_k^m) \prod_t \varphi(x_{tk}^{F,m}; F_t, (s_{tk}^{F,m})^2). \quad (9.4)$$

This is precisely an HMM normal-means smoothing problem. If a location has $a_{tk}^{F,m} = 0$, its emission likelihood is one: the chain still supplies information. A finite-SE-only solver needs an explicit missing-emission extension for that case; a large finite SE is only an approximation.

9.2 Emissions, forward/backward and moments

Suppress (m, k) and write x_t, s_t^2 . For state b , component c ,

$$e_{tbc} = \int \varphi(x_t; f, s_t^2) G_{bc}(df), \quad B_t(b) = \sum_c \rho_{bc} e_{tbc}. \quad (9.5)$$

For a signed Gaussian component, $e_{tbc} = \varphi(x_t; c_b, s_t^2 + v_c)$; for an atom, $e_{tbc} = \varphi(x_t; c_b, s_t^2)$. For a positive continuous component, put $v_* = (s_t^{-2} + v_c^{-1})^{-1}$, $m_* = v_*(x_t/s_t^2 + c_b/v_c)$, and use

$$e_{tbc}^+ = \varphi(x_t; c_b, s_t^2 + v_c) \frac{\Phi(m_*/\sqrt{v_*})}{\Phi(c_b/\sqrt{v_c})}. \quad (9.6)$$

Compute state smoothing using scaled or log-space versions of

$$f_1(b) = \alpha_b B_1(b), \quad f_t(b) = B_t(b) \sum_a f_{t-1}(a) T_{ab}, \quad (9.7)$$

$$g_{P_m}(b) = 1, \quad g_t(a) = \sum_b T_{ab} B_{t+1}(b) g_{t+1}(b). \quad (9.8)$$

After normalization,

$$\Gamma_t(b) \propto f_t(b) g_t(b), \quad (9.9)$$

$$\Xi_t(a, b) \propto f_{t-1}(a) T_{ab} B_t(b) g_t(b), \quad t \geq 2, \quad (9.10)$$

$$W_{tbc} = \Gamma_t(b) \frac{\rho_{bc} e_{tbc}}{B_t(b)}. \quad (9.11)$$

Each Γ_t , Ξ_t , and W_t sums to one over its state indices. Within a component, the conditional effect posterior is a Gaussian $\mathcal{N}(m_*, v_*)$, its positive truncation, or the corresponding atom. For the positive truncation let $z = m_*/\sqrt{v_*}$ and $r(z) = \varphi(z; 0, 1)/\Phi(z)$. Then

$$\mathbb{E}(F_t \mid b, c, x) = m_* + \sqrt{v_*} r(z), \quad \mathbb{E}(F_t^2 \mid b, c, x) = m_*^2 + v_* + m_* \sqrt{v_*} r(z). \quad (9.12)$$

Weight these moments by W_{tbc} . Stable tail evaluation is necessary when z is very negative. Reflection gives the negative model's moments. If requested, $\text{lfsr}_t = \min\{q(F_t \leq 0), q(F_t \geq 0)\}$; an atom at zero is in both tails, so in either one-sided model this equals $q(F_t = 0)$.

9.3 What can be reused

For fixed HMM parameters, the existing forward/backward and emission-moment machinery solves (9.4). Learning HMM parameters by increasing its normal-means marginal likelihood, while optimizing the corresponding factor posterior and holding the rest fixed, can serve as an EB block update. This requires an increase in the relevant common objective; it is not a guarantee for every heuristic initialization, center move or pruning proposal. State-support changes need particular care (Section 12.3).

10 Learning parameters and evaluating one objective

10.1 Neural gates

For each component, hold the variational distribution fixed and maximize

$$\sum_i [(1 - \gamma_{iu})\mathbb{E} \log(1 - \pi_{\theta_u}(P_{iu})) + \gamma_{iu}\mathbb{E} \log \pi_{\theta_u}(P_{iu})] + \log p(\theta_u). \quad (10.1)$$

Omit the last term for unpenalized EB. This is expected Bernoulli cross-entropy with soft responses and uncertain predictors. Sampling parent values supplies a stochastic estimate of its gradient. During this parameter step, q is fixed. During a *loading* step, dependence of all relevant child probabilities on that loading must be retained. These are distinct operations. The current isolated normal-means network fit does not by itself implement this global alternating optimization.

10.2 Global Gaussian slab mean and variance

Let $m_{iu} = \mathbb{E}V_{iu}$, $M_{iu} = \mathbb{E}V_{iu}^2$, and $G_u = \sum_i \gamma_{iu} > 0$. Without parameter priors, the M-step is

$$\mu_u^{\text{new}} = \frac{\sum_i m_{iu}}{G_u}, \quad d_u^2 = \frac{\sum_i \{M_{iu} - 2\mu_u^{\text{new}}m_{iu} + (\mu_u^{\text{new}})^2\gamma_{iu}\}}{G_u}. \quad (10.2)$$

These are posterior latent moments; there is no subtraction of the normal-means observation variance here. Formula (10.2) is for a free Gaussian slab, not for a truncated slab or the constraint $d = \omega|\mu|$. Those cases require their own expected log-density optimization, including normalizing constants. If $G_u = 0$, slab parameters are unidentified and an explicit convention or prior is needed. A variance floor changes the domain of the optimization and should be stated.

10.3 HMM parameters on fixed component supports

For one chain, standard expected-count maximizers are

$$\alpha_b^{\text{new}} = \Gamma_1(b), \quad T_{ab}^{\text{new}} = \frac{\sum_{t=2}^{P_m} \Xi_t(a, b)}{\sum_{t=2}^{P_m} \Gamma_{t-1}(a)}, \quad \rho_{bc}^{\text{new}} = \frac{\sum_t W_{tbc}}{\sum_t \Gamma_t(b)}. \quad (10.3)$$

Only update parameters that are free. Fixed initial probabilities, tied mixtures, forbidden transitions and zero-occupancy states require corresponding constraints or conventions. In particular, if α is constrained to be the stationary distribution of the *current* T , it cannot be optimized independently with these formulas. The package's default initial probabilities are fixed during an individual fit after their initialization. Dirichlet MAP parameters add $\alpha^{\text{prior}} - 1$ pseudo-counts on allowed entries where an interior MAP update exists.

10.4 Noise and likelihood terms

Under (5.1),

$$\mathcal{E}_{it}^m = \mathbb{E} \left(Y_{it}^m - \sum_k L_{ik}^m F_{tk}^m \right)^2 = \left(Y_{it}^m - \sum_k m_{ik}^{L,m} m_{tk}^{F,m} \right)^2 + \sum_k \left(M_{ik}^{L,m} M_{tk}^{F,m} - (m_{ik}^{L,m} m_{tk}^{F,m})^2 \right). \quad (10.4)$$

For a shared precision in an observed-entry group $\mathcal{G}$, its unpenalized maximizer is

$$\tau_{\mathcal{G}}^{\text{new}} = \frac{|\mathcal{G}|}{\sum_{(i,t) \in \mathcal{G}} \mathcal{E}_{it}^m}. \quad (10.5)$$

This presumes positive total residual and nonempty groups; unrestricted entry-specific variance estimation is generally not an appropriate model. With a Gamma prior of shape a_0 , rate b_0 , an independent variational factor is Gamma with shape $a_0 + |\mathcal{G}|/2$, rate $b_0 + \sum \mathcal{E}_{it}/2$. A Gibbs noise update uses realized squared residuals instead. Known precisions are not re-estimated.

The likelihood contribution to the ELBO is

$$\mathbb{E} \log p_m(Y^m | \cdot) = \frac{1}{2} \sum_{(i,t) \in \Omega_m} [\log \tau_{it}^m - \log(2\pi) - \tau_{it}^m \mathcal{E}_{it}^m] \quad (10.6)$$

for fixed precisions. Use $\mathbb{E} \log \tau$ and $\mathbb{E} \tau$, respectively, under an independent variational precision factor. For a loading mixture, $H(q_u) = -(1 - \gamma_u) \log(1 - \gamma_u) - \gamma_u \log \gamma_u + \gamma_u h(q_{u,1})$, on the disjoint spike/slab reference space. The tilted slab entropy can be evaluated from $\log q_{u,1}(v) = \log \varphi(v; m_0, v_0) + D_u(v) - \log I_0$. The HMM KL can be evaluated from its prior, normalized posterior and normal-means evidence; one need not force feature-position independence.

11 Simpler cases and a stronger variational alternative

11.1 Strict binary loadings: a closed-form scalar probability

Consider the different model $V_u \in \{0, 1\}$ with $g_u(1 | P_u) = \pi_u(P_u)$. Under independent Bernoulli variational factors, set $\gamma_u = q(V_u = 1)$. Then (5.8) gives

$$\boxed{\text{logit } \gamma_u = b_u - \frac{1}{2} a_u + \mathbb{E} \text{logit } \pi_u(P_u) + D_u(1) - D_u(0).} \quad (11.1)$$

The $-a_u/2$ term is essential because $1^2 - 0^2 = 1$. There is no continuous integral. Parent expectations may still require sums or approximation. For a small number of binary parents they can be enumerated exactly. The exact Gibbs probability uses fixed parents and children in the same expression. This is an excellent reference model for the notebook's binary truth, but it is not the package's continuous `gbinary` or `CGB`.

11.2 Linear-logistic CGB: a log-concave slab

Suppose each child gate has a linear logit in the candidate loading, $\pi_c(P_c[v]) = \text{sigmoid}(\beta_{cu}v + \zeta_c)$, where ζ_c depends on the other parents. Then under the same mean-field assumptions,

$$D'_u(v) = \sum_{c \in \mathcal{C}_u} \beta_{cu} \{ \gamma_c - \mathbb{E} \pi_c(P_c[v]) \}, \quad (11.2)$$

$$D''_u(v) = - \sum_{c \in \mathcal{C}_u} \beta_{cu}^2 \mathbb{E} \{ \pi_c(P_c[v]) (1 - \pi_c(P_c[v])) \} \leq 0. \quad (11.3)$$

Hence the CGB slab log posterior has second derivative $-1/v_0 + D_u''(v) < 0$. It is strictly log-concave and has one mode. One-dimensional quadrature or a suitable slice sampler is therefore simpler than for arbitrary neural logits. This result assumes parent-independent Gaussian slabs and linear logits; it is not a claim about the existing ReLU networks. Interactions can be introduced carefully, but an arbitrary nonlinear function of the target need not retain concavity.

For a child with positive β_{cu} , an inferred activation exceeding the gate's predicted activation makes $D_u'(v) > 0$ for that contribution. It pushes the parent upward. This gives a direct algebraic interpretation of RNA-to-ATAC posterior information.

11.3 Keep all loadings of a cell together

A richer alternative replaces the product of scalar loading factors by $\prod_i q_i(A_i, R_i)$. It can retain both within-modality and cross-modality posterior dependence. With all other blocks fixed, define

$$H_i^m = \sum_{t:(i,t) \in \Omega_m} \tau_{it}^m \mathbb{E}(F_{t,:}^m (F_{t,:}^m)^\top), \quad (11.4)$$

$$h_i^m = \sum_{t:(i,t) \in \Omega_m} \tau_{it}^m Y_{it}^m \mathbb{E} F_{t,:}^m. \quad (11.5)$$

Here $F_{t,:}^m$ is understood as a column vector. The unrestricted row coordinate is

$$q_i^*(a, r) \propto p(a, r \mid \theta) \exp \left\{ -\frac{1}{2} a^\top H_i^A a + (h_i^A)^\top a - \frac{1}{2} r^\top H_i^R r + (h_i^R)^\top r \right\}. \quad (11.6)$$

For strict binary loadings this can be normalized by enumerating $2^{K_A+K_R}$ states per cell. With five factors in each modality this is 1,024 states per cell; feasibility still depends on network evaluation and the number of cells and iterations. Continuous CGB loadings make the row integral multivariate and substantially harder.

Row dependence changes the factor pseudo-data. In particular, replace the factor numerator by

$$b_{tk}^{F,m} = \sum_{i:(i,t) \in \Omega_m} \tau_{it}^m \left[Y_{it}^m \mathbb{E} L_{ik}^m - \sum_{h \neq k} \mathbb{E}(L_{ik}^m L_{ih}^m) \mathbb{E} F_{th}^m \right]. \quad (11.7)$$

The denominator is $\sum_i \tau_{it}^m \mathbb{E}[(L_{ik}^m)^2]$, using the second moment rather than $(\mathbb{E} L_{ik}^m)^2$. The HMM form remains intact, but the existing mean-field residual formula is insufficient: within-row cross moments are needed. Likewise the residual second moment must be evaluated with those cross moments. Conditional on fixed F , exact binary row Gibbs uses the same construction with realized factor products in place of their expectations.

12 Qualifications needed for a coherent implementation

12.1 Covariate standardization

The package's `standard_scale` computes a column mean and standard deviation over the current input rows. If these statistics are recomputed as functions of latent loading columns *inside the probability model*, changing one A_{ik} changes transformed covariates for other cells. Later conditional priors in those cells then enter its Markov blanket too. The cell-local equations above would be incomplete.

For a first joint specification, choose fixed offsets/scales, for example from an initialization, and regard them as known transformations. Alternatively define a model with explicit transformation parameters and optimize them under one objective. Recomputing transformations between steps without accounting for their effect does not preserve a fixed joint inference target.

12.2 The current sparsity penalty is not automatically a parameter prior

With fixed covariates the CGB loss adds $(\lambda - 1) \sum_i \log(1 - \pi_{iu}(X_i))$ to the log-likelihood criterion. If the covariates become latent parents, that expression depends on latent variables, not only on neural parameters. One possible explicitly regularized variational criterion is

$$\mathcal{L}_{\text{reg}}(q, \Theta) = \mathcal{L}(q, \Theta) + (\lambda - 1) \sum_u \mathbb{E}_q \log(1 - \pi_u(P_u)). \quad (12.1)$$

Its loading-coordinate update must include the added terms whenever the target is a parent. It is not the ELBO of the unpenalized joint model. To interpret the same factor as a new normalized tilted joint density, its normalizing constant must be included; this generally depends on parameters and affects their learning. Simply multiplying each child’s conditional by a function of its parents and renormalizing that conditional cancels the function, so it does not give the intended regularization.

A clean initial derivation therefore uses $\lambda = 1$, with optional proper priors or explicit regularizers on *parameters alone*. HMM Dirichlet parameter priors have a different, direct expected-count interpretation. The same numerical penalty value across different solvers need not mean the same statistical regularization.

12.3 Ordering, rescaling, support changes and pruning

For each modality, $L_k \mapsto cL_k, F_k \mapsto F_k/c$ preserves the fitted matrix for $c \neq 0$. It need not preserve the complete prior or the joint objective. Prior parameters, neural input transformations and HMM supports must be transformed consistently before treating rescaling as a mere reparameterization. Fix a scale convention for interpretation and initialization.

Order matters in (3.4). Pruning or permuting a component changes the identities and dimensions of later predictors. Such changes require model reconstruction and objective evaluation, not only deletion of one loading column. A deterministic neural conditional can also create nearly hard dependencies and poor single-coordinate mixing; bounded gate probabilities and row-block inference can help, but change the approximation or admissible parameter space and should be specified.

Moving the location of a point mass while holding its old posterior fixed can make the expected prior log density minus infinity. One should not apply an ordinary fixed-posterior M-step blindly to moving atoms. Keeping HMM atoms and grids fixed is the simplest derivation. Jointly refitting a factor posterior and its HMM under a changed support can instead be considered as an EB block proposal, assessed using its collapsed marginal likelihood and the consistent joint objective. The distinction is between a valid evaluated block proposal and an assumed fixed-support EM update.

12.4 A mathematically defined iteration

For the unpenalized fixed-rank, fixed-scaling model, one possible schedule is:

1. Initialize all loading distributions, factor HMM posteriors and parameters.
2. Update each ATAC loading using (5.8), including later ATAC and RNA children; retain its spike and slab representation.
3. Update each RNA loading, including its later RNA children.
4. Update each factor sequence by (9.4) and forward/backward.

5. Update shared loading-prior, HMM and noise parameters using the common objective. A generalized EM step must increase that objective, even if not maximizing it completely.
6. Evaluate the *single* objective (5.3) at the current state. Check convergence and numerical accuracy; repeat if needed.

Other scan orders are possible. Exact coordinate steps optimize the same objective; local optima and rates of convergence can still depend on order and initialization. Keeping old per-factor KL values after changing their parents does not evaluate the current joint objective. A Gibbs version samples full conditionals instead and targets a posterior rather than an ELBO.

13 Scientific scope and evaluation

13.1 What the construction could provide

1. **Bidirectional use of paired data.** RNA can resolve ambiguous ATAC loadings and ATAC can resolve ambiguous RNA loadings under one directed prior. A separate reverse prior is not required.
2. **Different ranks and modality-specific programs.** This model does not force $A = R$ or equal ranks. It models relationships among activity patterns while preserving modality-specific feature factors.
3. **A precise extension of EBNM modularity.** The ordinary Gaussian normal-means reduction remains, and the HMM factor block remains reusable. Equation (8.10) identifies exactly what a new loading solver must add.
4. **A testable objective.** The derivation separates a joint-model inference algorithm from a useful but uncalibrated covariate-exchange heuristic. It specifies what must be monitored and which terms a proposed shortcut drops.

This makes the derivation useful before investing in a general neural solver. It does not establish novelty, identifiability, causal direction or guaranteed improvement. EBMF’s normal-means reduction is established work [5]; covariate-modulated priors are the cEBMF setting [6]. The present note derives their ordered joint extension for this project.

13.2 Why the supplied simulation is informative

The noiseless RNA profiles distinguish the four binary ATAC states. For the three RNA blocks at positions 200–249, 500–549 and 700–749 (Python indexing), the mapping is

ATAC loading 1	ATAC loading 2	RNA block 1	RNA block 2	RNA block 3
0	0	1	1	0
0	1	1	1	1
1	0	0	1	1
1	1	1	0	1

Thus RNA has information about the ATAC state that could improve ATAC inference when its own observations are noisy. The two independent noise realizations make combining their evidence sensible under the correctly specified joint model. The four one-hot RNA loading columns used to generate the data do not force fitted RNA rank four: the noiseless matrix has rank three, and a three-block factorization is another representation.

Finite-sample estimation, misspecified gates, incorrect pairing and local optima can make sharing harmful. Under a correct known model, conditioning on more data cannot increase the expected Bayes squared-error risk of a posterior mean; that statement does not guarantee improvement in every realized dataset or for fitted empirical Bayes/variational approximations. More concretely, for a square-integrable target U , the conditional-expectation identity gives

$$\mathbb{E}\{(U - \mathbb{E}[U | Y_A])^2\} - \mathbb{E}\{(U - \mathbb{E}[U | Y_A, Y_R])^2\} = \mathbb{E}\{(\mathbb{E}[U | Y_A, Y_R] - \mathbb{E}[U | Y_A])^2\} \geq 0. \quad (13.1)$$

13.3 The most informative first comparison

A small strict-binary version with exact enumeration can validate the posterior-feedback equations without neural or quadrature uncertainty. A linear-logistic CGB version can then isolate the continuous-loading problem while retaining a log-concave scalar slab. A flexible neural version becomes more informative after those reference cases are understood.

Useful comparisons include independent fits, each fixed one-direction fit, the covariate-exchange heuristic, and the joint-target method. Evaluate both modalities, using held-out observations excluded from *every* preprocessing and fitting stage that could leak those values. In simulations also compare against the noiseless matrices and assess loading uncertainty where a common scale/order is defined. Check shuffled cell pairing, multiple initializations, different scan orders, and weak or absent cross-modality dependence. Section 14 places these comparisons in the existing literature; the derivation alone does not establish methodological novelty.

14 Comparison with other multi-omics methods

14.1 Scope and notation

This section compares the proposed model with representative multi-omics methods, using the literature reviewed through 15 September 2026. It is a comparison of statistical structure and inferential requirements, not an empirical performance ranking or an exhaustive novelty review. Throughout, L^m denotes sample loadings and F^m denotes feature factors; some cited methods use the opposite terminology.

Three questions must be separated: which finite signal matrices can be represented, which joint distributions of sample programs can be modeled, and how accurately those distributions can be learned and their posteriors approximated. A more flexible conditional family need not improve finite-sample prediction, calibration, or computation.

For reference, a shared-score linear predictor has the form

$$Y^m = ZW_m^\top + E^m, \quad (14.1)$$

whereas a shared/private decomposition can be written schematically as

$$Y^m = SU_m^\top + V_mB_m^\top + E^m. \quad (14.2)$$

Here Z, S are sample-score matrices shared across views and V_m contains view-specific sample scores. The proposed formulation instead uses $Y^A = AF_A^\top + E^A$, $Y^R = RF_R^\top + E^R$, and the normalized conditional prior (3.4). These schematic matrix equations do not identify methods' complete model classes: their priors, likelihoods, rank constraints, and identifiability restrictions can differ.

14.2 Population priors and the relation to TwinEB

The closest comparison for flexible empirical-Bayes prior learning is TwinEB, introduced by Salehi, Nazaret, Shah and Blei [1]. TwinEB learns mixture priors on the entire row and column latent vectors and fits these priors jointly with a variational posterior. The paper develops Gaussian and Poisson matrix factorization. Its mixture-of-vectors construction must be distinguished from a product of independent scalar mixture priors: a shared component label can induce dependence among coordinates of a latent vector.

For the Gaussian formulation, write a diagonal-component row mixture as

$$p_{\text{pop}}(z) = \sum_{h=1}^H \omega_h \prod_{k=1}^K \varphi(z_k; m_{hk}, s_{hk}^2). \quad (14.3)$$

The scales in the paper are parameterized per component and coordinate (Appendix B). Its mean-field posterior approximation does not imply independence under the prior. For example, with $\bar{m} = \sum_h \omega_h m_h$,

$$\text{Cov}(z) = \sum_h \omega_h \text{diag}(s_h^2) + \sum_h \omega_h (m_h - \bar{m})(m_h - \bar{m})^\top.$$

The second term may have nonzero off-diagonal entries. Thus dependence between latent programs is not, by itself, a capability absent from TwinEB.

An exact relationship at the level of the loading prior. The vector-mixture form implies the autoregressive conditionals

$$p_{\text{pop}}(z_k \mid z_{<k}) = \sum_h \tilde{\omega}_{hk}(z_{<k}) \varphi(z_k; m_{hk}, s_{hk}^2), \quad (14.4)$$

$$\tilde{\omega}_{hk}(z_{<k}) = \frac{\omega_h \prod_{j<k} \varphi(z_j; m_{hj}, s_{hj}^2)}{\sum_r \omega_r \prod_{j<k} \varphi(z_j; m_{rj}, s_{rj}^2)}. \quad (14.5)$$

To prove the result, let $C_k(z_{\leq k}) = \sum_h \omega_h \prod_{j \leq k} \varphi(z_j; m_{hj}, s_{hj}^2)$, with $C_0 = 1$. The conditional in (14.4) equals C_k/C_{k-1} . Multiplying over k telescopes to C_K , which is (14.3).

Accordingly, a conditional Gaussian-mixture framework that permits the specific gates (14.5) contains this row-prior family exactly. An unrestricted conditional head can also vary the component means and variances with preceding coordinates, and incorporate observed covariates. However, an arbitrary finite ReLU network does not necessarily implement the log-quadratic Bayes gates exactly. A one-slab CGB is also a much smaller family than an unrestricted vector mixture. The result is a distributional inclusion under explicit assumptions, not a blanket inclusion claim for every registry configuration.

Any joint density admits an autoregressive factorization by the chain rule. Moreover, increasing the number of Gaussian mixture components already gives highly flexible continuous population models. The proposed ordering therefore does not establish that TwinEB cannot represent nonlinear dependence. The empirical question is whether the chosen conditional parameterization represents and learns scientifically relevant dependencies more efficiently at feasible sample sizes and computational budgets. Exact atomic probabilities are an additional modeling choice; narrow continuous Gaussians approximate concentration near zero without assigning positive probability to zero itself.

Different structural assumptions. The present model explicitly separates modality-specific latent programs and specifies

$$p(A_i, R_i \mid X_i) = p(A_i \mid X_i)p(R_i \mid A_i, X_i)$$

through within- and between-modality conditional distributions. This allows external covariates to modify activation probabilities, amplitudes or scales at particular nodes. All such conditional factors belong to one normalized joint model; upstream updates include the downstream evidence described in (7.10).

TwinEB, as formulated in the cited paper, uses population priors for a single matrix. A multi-view extension could concatenate loading vectors and specify a suitable block observation model. That is a legitimate comparator construction, but it should be identified as an extension and implemented with the same missing-data likelihood and evaluation protocol rather than attributed to the published model.

The feature-factor assumptions differ as well. TwinEB’s column prior is a mixture on an entire feature’s latent vector; it can relate factor coordinates within that vector. The present HMM instead relates adjacent feature positions within each factor and is independent across factors. It is not a strict superset of the TwinEB column-prior model. Similarly, the current Gaussian observation implementation does not subsume the paper’s Poisson formulation.

Inference and the proposed contribution. Both approaches use learned distributions to regularize matrix factorization. TwinEB optimizes prior and mean-field variational parameters through an ELBO. The reference solver here uses analytic normal-means proposals with a child-likelihood Metropolis correction, sequence sampling for HMM factors, and optional partial Monte Carlo empirical-Bayes learning. Section 7.5 supplies an alternative augmented variational formulation; it has not been established to be faster or more accurate than TwinEB’s inference.

A defensible methodological contribution is therefore a modular conditional-prior factorization framework with explicit sparse or positive distributions, fixed side information, ordered feature priors, and joint uncertainty propagation across partially observed modalities. The distributional decomposition and the Metropolis identity are established mathematics. Novelty would rest on a useful model construction, effective inference and evidence that these structural choices improve a defined scientific task.

Comparisons needed to support this positioning. Include TwinEB alongside independent cEBMF and the multi-omics comparators below. On a common Gaussian single-matrix benchmark, compare vector population mixtures and autoregressive mixtures at matched ranks and documented parameter and tuning budgets. For multi-view data, report any required extension of the competitor. Separately ablate the hierarchy, nonlinear means and scales, exact spikes, exponential slabs, fixed covariates and HMM smoothing. This distinguishes prior expressiveness, regularization, modality coupling and the inference algorithm.

Evaluation should include held-out observations, missing-modality prediction, predictive calibration, and computation at matched accuracy. For samplers, report effective sample sizes and convergence diagnostics rather than acceptance rates alone. For VI, assess the effect of its dependence assumptions against small reference problems. No result in this derivation establishes empirical superiority over TwinEB or the other methods.

14.3 Structural comparison

Table 2 summarizes the cited formulations. A mechanism absent from a standard formulation may be possible in an extension. In particular, neither modality-specific programs nor joint

borrowing of information is unique to the proposed hierarchy.

Table 2: Representative multi-omics methods and their relation to the proposed conditional loading model. Dates refer to the cited papers.

Method	Native model structure	Relation to this derivation
TwinEB (2024)	Learns mixture population priors on complete row and column latent vectors, with joint variational empirical-Bayes estimation.	Vector mixtures already capture coordinate dependence. Explicit conditional covariates, modality structure and HMM adjacency are distinct choices; Section 14.2 gives the precise relationship [1].
JIVE (2013); AJIVE (2018)	Decompose joint and individual low-rank sample subspaces. AJIVE uses subspace angles to estimate shared variation.	Original formulations describe linear shared structure rather than learned nonlinear conditional distributions between distinct sample programs [8, 9].
MOFA+ (2020)	Shared sample scores and view-specific feature weights, with shrinkage, groups, missing data, and several observation likelihoods.	Already supports shared, view-specific and subset-active factors. The standard score-prior construction differs from the autoregressive conditional loading prior (3.4) [7].
ProJIVE (2026)	Probabilistic joint and individual factorization, fitted by maximum likelihood with EM.	A probabilistic extension of JIVE already exists; probability modeling alone is not a distinguishing contribution [14].
SLIDE (2019); BASS (2016)	Components shared across subsets of views. SLIDE uses structured sparsity; BASS uses Bayesian structured shrinkage.	Partial sharing and sparse/dense program structure are established capabilities, distinct from conditional dependence among programs [10, 11].
BIDIFAC+ (2022); EV-BIDIFAC (2024)	Low-rank modules shared across subsets of linked row and column groups; EV-BIDIFAC uses empirical variational Bayes shrinkage.	Close empirical Bayes integration baselines, including missing-data imputation. The proposed additional mechanism is learning conditional program distributions [12, 13].
SIFA (2017)	Joint and individual factors partially informed by observed auxiliary covariates through nonparametric functions.	Nonlinear covariate-informed factors already exist. Uncertain latent parents and the required posterior feedback distinguish the present inference problem [17].
MuVI (2023); SOFA (2026)	MuVI incorporates feature-set information through structured shrinkage. SOFA links guided factors to sample-level guide variables and retains unguided factors.	Relevant comparisons when biological feature sets or sample annotations are supplied. They address other uses of prior information [18, 19].
MEFISTO (2022)	Gaussian-process structure on sample factors along continuous covariates such as time and space.	Sample-axis covariance differs from the feature-axis HMM in Section 9. Structured factor priors themselves are not new [20].
CAVACHON (2024 preprint)	A supplied modality DAG organizes a hierarchical VAE with mixture innovation latents and ancestor-aware decoders.	Close prior art for biological hierarchy and conditional independence; its decoder construction differs from normalized scalar loading priors with linear feature reconstruction [15].

Continued on the next page

Table 2 continued

Method	Native model structure	Relation to this derivation
HALO (2025)	RNA/ATAC representations linked through learned mappings and causal constraint losses; an interpretable nonlinear additive decoder.	Hierarchical RNA–ATAC analysis and interpretable decoding already exist. Its objective differs from the normalized joint-prior construction here [16].
MultiVI (2023); GLUE (2022)	MultiVI integrates paired and unpaired RNA/ATAC data with a multimodal VAE. GLUE links modality embeddings through a feature guidance graph.	Important single-cell comparators with different information requirements and observation models. The present paired Gaussian model has narrower scope [21, 22].

14.4 Observed side information and uncertain latent parents

The model can combine fixed information x_i^A, x_i^R with latent programs:

$$\begin{aligned} A_{ik} &| A_{i,<k}, x_i^A \sim g_k^A(\cdot | A_{i,<k}, x_i^A), \\ R_{ij} &| R_{i,<j}, A_i, x_i^R \sim g_j^R(\cdot | R_{i,<j}, A_i, x_i^R). \end{aligned} \quad (14.6)$$

This makes explicit the fixed covariates suppressed in most earlier equations. In (6.2), replace every $g_u(v | P_u)$ by $g_u(v | P_u, x_u)$, and every child density by $g_c(V_c | P_c[v], x_c)$. Average over uncertain latent inputs, while holding the observed x ’s fixed. The fixed inputs can modulate child feedback and do not disappear from those density evaluations.

EBMF’s reduction to normal-means problems is established [5], and flexible priors informed by fixed side information are already part of cEBMF [6]. Thus the added inferential problem is not simply concatenating more covariates. It is that changing a latent loading changes other factors’ priors, requiring both uncertain-parent expectations and child contributions under one objective. The same framework can condition on observed context and latent programs simultaneously; it need not choose between them.

14.5 Comparison with hierarchical neural models

CAVACHON’s generative equations use Gaussian-mixture innovation variables and feed ancestor representations into the observation decoder [15]. In the present construction, the hierarchy acts on scalar loading distributions, while $F^m(L_i^m)^\top$ retains a linear additive feature representation. These architectures impose different assumptions; neither comparison alone proves greater distributional capacity.

HALO’s displayed generative specification uses Gaussian latent priors, with additional mapping and causal-constraint terms in its training objective (Methods, Equations 17–24) [16]. It should therefore not be identified with exactly the normalized conditional-prior model in (3.4). Conversely, hierarchical RNA–ATAC modeling and interpretability alone cannot distinguish this proposal from HALO.

The potential contribution is a particular statistical model and inference construction: normalized conditional program distributions, explicit additive feature factors, and EBNM-based updates across multiple prior families. Accurate uncertainty, computational advantages and biological usefulness remain empirical questions.

14.6 Reconstruction capacity versus conditional dependence

Shared-score representation of any fitted pair. Let $M^A = AF_A^\top$, $M^R = RF_R^\top$, and set

$$Z = [A \ R], \quad W_A = [F_A \ 0], \quad W_R = [0 \ F_R]. \quad (14.7)$$

Then $M^A = ZW_A^\top$ and $M^R = ZW_R^\top$. A shared-score representation with at most $K_A + K_R$ columns can therefore reproduce these finite fitted matrices exactly. This is a statement about matrix parameterization, not equality of their latent prior distributions or of the full model classes. It rules out a universal reconstruction advantage based solely on separate loading matrices.

Linear Gaussian hierarchy as shared/private structure. For column-vector loadings, suppose $a \sim \mathcal{N}(0, \Sigma_A)$ and $r = Ba + \eta$, with $\eta \sim \mathcal{N}(0, \Sigma_\eta)$ independent of a . The observation equations become

$$y^A = F_A a + \epsilon_A, \quad y^R = F_R B a + F_R \eta + \epsilon_R. \quad (14.8)$$

Thus a supplies shared variation and η supplies RNA-specific variation. For positive-definite covariance matrices, whitening gives independent standard Gaussian scores with transformed feature weights. At the observation-distribution level this is shared/private linear Gaussian structure, although method-specific sparsity or orthogonality constraints can still differ.

Nonlinear dependence. Let A_1, A_2 be independent Bernoulli(1/2) variables and $R = A_1 \text{ xor } A_2$. Then $\mathbb{E}R = 1/2$ and $\text{Cov}(A_1, R) = \text{Cov}(A_2, R) = 0$. The optimal affine predictor from these two inputs is 1/2, with population squared-error risk 1/4; the nonlinear oracle

$$\mathbb{E}(R \mid A_1, A_2) = A_1 + A_2 - 2A_1A_2 \quad (14.9)$$

has zero risk. A neural CGB gate can approximate this relationship as its activation probabilities approach zero/one and its active slab concentrates near one. It is not exactly a binary prior at finite positive slab variance.

This example establishes a limitation of affine prediction using the specified two inputs. An expanded basis containing A_1A_2 represents the mean with a linear readout, and non-Gaussian latent models can have nonlinear posterior predictors. These calculations are not fitted MOFA+ or JIVE error rates and do not prove that every configuration of those methods fails on XOR.

Conditional variance can be useful without a mean association. Suppose $R \mid A = a \sim \mathcal{N}(0, v(a))$ and $T = R + \varepsilon$ with independent $\varepsilon \sim \mathcal{N}(0, s^2)$. Then $\mathbb{E}(R \mid A) = 0$, but

$$\mathbb{E}(R \mid T, A = a) = \frac{v(a)}{v(a) + s^2} T, \quad \text{Var}(R \mid T, A = a) = \frac{v(a)s^2}{v(a) + s^2}. \quad (14.10)$$

ATAC can inform the amount of RNA shrinkage even without predicting its conditional mean from ATAC alone. This is relevant to the centered conditional scale-mixture families in Section 6.4. Predictive dispersion and calibration should therefore supplement cross-modal mean prediction in a benchmark.

14.7 Implications for inference and empirical comparison

Full joint inference allows evidence from either modality to update the other, including when the generative prior is directed. This is not unique to the present model: shared latent-variable

models also borrow information from all contributing views. The relevant question is whether the proposed conditional distributions provide better assumptions for a particular scientific problem.

A dense graph with unrestricted conditionals can express any joint law by the chain rule in different orders. It does not identify biological causality. Sparse parents, restricted conditional families and anchored factor definitions can make a hierarchy more informative, but their assumptions and sensitivity require assessment. Scale, permutation, sign and possible rotation ambiguities must be accounted for when interpreting dependencies between estimated programs.

An informative comparison should include independent cEBMF, one-way plug-in conditioning in each direction, within-modality hierarchy only, a linear conditional model, and the joint-target updates derived here. A two-stage nonlinear map between independently fitted loadings helps separate the benefit of flexible regression from that of joint inference. External comparators should include TwinEB and both linear/empirical Bayes methods and hierarchical neural models in Table 2.

Evaluate held-out prediction and calibration separately for both modalities, with ranks and tuning budgets treated fairly. Vary noise asymmetry, nonlinear gating, variance-only dependence, incorrect or absent dependence, and shuffled pairing. Hold feature priors fixed in loading-model ablations so that HMM smoothing is not mistaken for a benefit of the hierarchy. The HMM uses feature adjacency, whereas fixed genomic distances or sample coordinates are different inputs.

For new-sample prediction, exclude held-out observations from preprocessing, factor fitting, co-variate construction and conditional-network training. Separate entry-wise completion from entirely missing-modality prediction, and split by donors or studies when that is the unit of generalization. Compare predictions for the same observed quantity and units; raw ELBOs across different models or Gaussian transformed-data losses against count likelihoods are not direct performance comparisons. Original deterministic decompositions do not provide a predictive density without an additional model.

The present Gaussian observation model does not automatically inherit count-data, batch-correction or unpaired-integration capabilities from the cited methods. Likewise, the general coordinate identity and Metropolis rule are established tools [4, 3]; their application is not itself a novel general theorem. A methodological contribution would require a coherent implemented solver, evidence that its approximation is accurate, and a demonstrated advantage of the conditional-program assumptions in appropriate settings.

A Deriving the likelihood coefficients

For one observed entry, write the contribution excluding the target as $S_{-k} = \sum_{h \neq k} L_{ih} F_{th}$. Expanding gives

$$(Y - vF_{tk} - S_{-k})^2 = v^2 F_{tk}^2 - 2vF_{tk}(Y - S_{-k}) + (Y - S_{-k})^2. \quad (\text{A.1})$$

Under (5.1), $\mathbb{E}F_{tk}^2 = M_{tk}^F$ and $\mathbb{E}[F_{tk}(Y - S_{-k})] = m_{tk}^F(Y - \sum_{h \neq k} m_{ih}^L m_{th}^F)$. Multiplication by $-\tau/2$ and summation over observed features gives (5.6)–(5.7). For a factor update, interchange loading and factor roles. If loading columns are dependent within a row, $\mathbb{E}(L_{ik}L_{ih})$ cannot be factorized, yielding (11.7).

B Two-node binary illustration of posterior feedback

Let $A, R \in \{0, 1\}$, $A \sim \text{Bernoulli}(p)$, $R \mid A = a \sim \text{Bernoulli}(\rho_a)$. Suppose the ATAC likelihood alone contributes log odds λ_A for $A = 1$ against $A = 0$. Conditional on $R = r$,

$$\text{logit } p(A = 1 \mid R = r, Y_A) = \text{logit } p + \lambda_A + r \log \frac{\rho_1}{\rho_0} + (1 - r) \log \frac{1 - \rho_1}{1 - \rho_0}. \quad (\text{B.1})$$

For mean-field $q(R = 1) = \gamma_R$, replace r by γ_R to get the CAVI update. This is an expected *log* message. By contrast, the exact marginal posterior for A given RNA data with likelihoods $L_R(r) = p(Y_R \mid R = r)$ has the RNA likelihood ratio

$$\frac{(1 - \rho_1)L_R(0) + \rho_1 L_R(1)}{(1 - \rho_0)L_R(0) + \rho_0 L_R(1)}. \quad (\text{B.2})$$

The log of this ratio is generally not the mean-field expected-log term. This example distinguishes an exact scalar full conditional, an exact *marginal* posterior, and a variational coordinate update. They are three different quantities even though they arise from the same joint model.

C Implemented reference updates for the learned priors

The experimental implementation in `src/cebmf_torch/experimental/conditional.py` and `src/cebmf_torch/experimental/joint.py` targets the augmented latent posterior for fixed fitted parameters. It supports the eight scalar families in Section 6.4: CGB, both sharp CGB variants, CASH, LCASH, proportional-odds LCASH, EMDN and spiked EMDN. The existing neural-network classes are reused; normalized log weights, explicit slab scales, and a fixed pre-processing transform are exposed. The production `CEBMF.iter_once` method is unchanged and does not acquire these updates by replacing its covariate matrix.

C.1 An augmented normal-means proposal with exact atoms

Fix a sample and target node u , the other latent variables, and all parameters. Write

$$g_u(dv, h \mid P_u) = w_h G_h(dv), \quad \ell_u(v) = -\frac{1}{2}av^2 + bv.$$

For a Gaussian component, let μ_h and $d_h^2 > 0$ denote its mean and variance evaluated at the *current* parents. For an atomic component set $d_h^2 = 0$, and $G_h = \delta_{\mu_h}$. Atoms in the implemented scalar priors are fixed at zero. Define

$$t_h = 1 + ad_h^2, \quad V_h = d_h^2/t_h, \quad M_h = (\mu_h + bd_h^2)/t_h, \quad (\text{C.1})$$

$$\log I_h = -\frac{1}{2} \log t_h + \frac{-a\mu_h^2/2 + b\mu_h + b^2d_h^2/2}{t_h}. \quad (\text{C.2})$$

Direct integration gives $I_h = \int e^{\ell_u(v)} G_h(dv)$, including $I_h = e^{-a\mu_h^2/2 + b\mu_h}$ for an atom. Consequently the normalized proposal is

$$\tilde{w}_h = \frac{w_h I_h}{\sum_g w_g I_g}, \quad q_0(dv, h) = \tilde{w}_h \begin{cases} \mathcal{N}(dv; M_h, V_h), & d_h^2 > 0, \\ \delta_{\mu_h}(dv), & d_h^2 = 0. \end{cases} \quad (\text{C.3})$$

This expression avoids subtracting terms of size μ_h^2/d_h^2 when a slab is extremely narrow. If $a = b = 0$, it recovers the conditional prior exactly, without an artificial precision floor.

Keep each child's component B_c fixed during this move and define

$$D_u(v) = \sum_{c \in \text{ch}(u)} \{\log w_{cB_c}(P_c[v]) + \log G_{cB_c}(V_c \mid P_c[v])\}. \quad (\text{C.4})$$

If the child is in its fixed zero atom, its component log mass is zero on that event. If it is in a Gaussian slab, include its full log density. Sample $(v', h') \sim q_0$ and accept with probability

$$\alpha((v, h), (v', h')) = \min\{1, \exp[D_u(v') - D_u(v)]\}. \quad (\text{C.5})$$

Indeed the target is proportional to $q_0(dv, h)e^{D_u(v)}$; the forward and reverse proposal densities cancel, with respect to the same component-specific reference measures. This proves detailed balance for the augmented coordinate kernel. A deterministic sweep of these invariant kernels preserves the same joint distribution, although the whole sweep need not itself be reversible. It does not prove that a finite chain has mixed.

For CGB and fixed-component CASH variants, only child log weights vary with v . For EMDN and spiked EMDN, the child's Gaussian term also varies:

$$-\frac{1}{2} \log(2\pi d_{cB_c}^2(P_c[v])) - \frac{(V_c - \mu_{cB_c}(P_c[v]))^2}{2d_{cB_c}^2(P_c[v])}.$$

Dropping either contribution changes the stationary target. The implementation uses the full expression for every family, so the CGB simplification is obtained by cancellation rather than a different update path.

The augmented state includes a component label even for overlapping continuous components. Holding a child's label fixed is correct for this augmented sampler; replacing its density by a collapsed mixture in just some parent moves would require a separately justified partially collapsed scheme. Each node's own move jointly proposes its value and component, allowing transitions to and from the exact spike.

C.2 Matrix statistics and feature-factor updates

For modality m , let M_{it}^m indicate an observed entry, and define the residual excluding factor k using *realized* current values:

$$E_{it,-k}^m = Y_{it}^m - \sum_{\ell \neq k} L_{i\ell}^m F_{t\ell}^m.$$

For a loading coordinate,

$$a_{ik}^m = \tau_m \sum_t M_{it}^m (F_{tk}^m)^2, \quad b_{ik}^m = \tau_m \sum_t M_{it}^m E_{it,-k}^m F_{tk}^m.$$

For the feature-factor block, use

$$a_{tk}^m = \tau_m \sum_i M_{it}^m (L_{ik}^m)^2, \quad b_{tk}^m = \tau_m \sum_i M_{it}^m E_{it,-k}^m L_{ik}^m.$$

These are conditional sampling statistics, not the mean-field statistics that substitute posterior second moments. They support arbitrary missing entries as well as entire missing modality rows.

The HMM block update first integrates each component's value to obtain its emission likelihood, then samples the entire state sequence by forward filtering and backward sampling, and finally samples each component and factor value conditional on the sampled state. Positive HMM components use truncated-normal draws, including exact atoms; negative HMM components use reflection. With $a = b = 0$ at a position, its emission likelihood is exactly one, so adjacent positions can still inform the hidden state. Independent draws from smoothed marginal state probabilities would not sample the correct Markov path.

C.3 Parameter learning: what the prototype does and does not optimize

The initial independent **gbinary**/HMM fits determine factor counts, ordering and the initial HMM distributions. Rescaling a loading column by $1/c_k$ requires multiplying its feature factors, HMM means and HMM standard deviations by $c_k > 0$. These quantities are then fixed. The initial independent loading estimates initialize conditional-prior networks; they are not extra observations in the final likelihood.

The optional generalized Monte Carlo EM phase draws augmented loading states under the current joint model and approximately maximizes, for each node,

$$\hat{Q}_u(\theta_u) = \frac{1}{SN} \sum_{s=1}^S \sum_{i=1}^N \left[\log w_{uB_{iu}^{(s)}}(P_{iu}^{(s)}; \theta_u) + \log G_{uB_{iu}^{(s)}}(V_{iu}^{(s)} | P_{iu}^{(s)}; \theta_u) \right]. \quad (\text{C.6})$$

Parents, values and labels from each draw stay together in this expression; their empirical dependence is not broken into products of marginal means. No measurement variance is subtracted from these already latent draws. An Adam step is retained only if the best resulting parameters improve the same fixed-draw objective; otherwise the preceding parameters are restored. This safeguards $\hat{Q}_u$, not the true evidence: a finite, potentially poorly mixed Monte Carlo E-step is still approximate.

For CASH/LCASH only the gate parameters are learned, and their zero-centered scale grid is fixed. For CGB the global mean, slab variance and neural gate are trainable. For EMDN the weights, means and scales are learned jointly. Both sharp variants keep their initialized narrow slab scales fixed by default. If scale learning is requested, the actual effective variance is learned; ω then only changes initialization. Multiplying an unconstrained ML variance estimate by $\omega < 1$ at every M-step is not a maximization of (C.6). With an unrestricted base variance, the parameterization $d^2 = \omega\sigma^2$ cannot by itself enforce persistent sharpening.

The prototype explicitly bounds learned continuous standard deviations to a positive finite interval (default $[0.02, 20]$). This specifies an admissible subfamily and limits variance collapse; it does not remove the exact zero atom. The fixed ASH grid is supplied separately. The ordinary normalized model uses no extra sparsity penalty. The local EBNM penalty is *not* silently inserted into the joint target: Section 12.2 explains the additional latent-parent terms and, depending on the chosen interpretation, normalization it would require.

The observation noise, HMM parameters, graph, preprocessing and factor count are held fixed in this reference implementation. It therefore performs joint latent inference conditional on fitted parameters, with optional partial empirical-Bayes learning of the loading priors. It does not implement fully Bayesian parameter inference or joint evidence optimization over every HMM/noise parameter. The general parameter equations elsewhere in this note describe further extensions.

After learning, parameters are frozen and a new burn-in precedes retained draws. Reconstruction uses

$$\hat{\mathbb{E}}(L^m(F^m)^\top | Y_{\text{obs}}, X) = \frac{1}{S} \sum_s L^{m,(s)}(F^{m,(s)})^\top,$$

which generally differs from the product of marginal posterior means. Reported reconstruction standard deviations describe the latent signal, conditional on fitted parameters; they exclude new observation noise and hyperparameter uncertainty. Acceptance rates include accepted proposals that leave a zero value unchanged and are not convergence diagnostics.

D Partial overlap and fixed side information

Let I_A and I_R denote the observed sample IDs and $I = I_A \cup I_R$. All rows are explicitly aligned to I ; row positions are not assumed to match. Let X_i be finite fixed side information, with any transformation fitted once and frozen. The directed loading model is

$$p(A_i, R_i | X_i) = \prod_{k=1}^{K_A} g_k^A(A_{ik} | A_{i,<k}, X_i) \prod_{j=1}^{K_R} g_j^R(R_{ij} | A_i, R_{i,<j}, X_i). \quad (\text{D.1})$$

Conditional on the masks, the observed-data target is

$$p(A, R, F | Y_{\text{obs}}, X) \propto p(F) \prod_{i \in I} p(A_i, R_i | X_i) \times \prod_{m \in \{A, R\}} \prod_{(i,t): M_{it}^m=1} \varphi\left(Y_{it}^m; \sum_k L_{ik}^m F_{tk}^m, \tau_m^{-1}\right). \quad (\text{D.2})$$

This treats the observation pattern as ignorable for the modeled measurements, conditional on the supplied information. An informative measurement-selection mechanism would need its own model.

For an RNA-only row and fixed factors and parameters,

$$p(R_i | Y_i^R, X_i) \propto p(Y_i^R | R_i, F^R) \int p(R_i | A_i, X_i) p(A_i | X_i) dA_i. \quad (\text{D.3})$$

Thus the *marginal prior* for its RNA loadings is a function of fixed side information and the learned population relationship, integrating out unknown ATAC loadings. At a particular RNA coordinate it also conditions on earlier RNA coordinates as prescribed by the DAG. This is generally not the same distribution obtained by deleting the ATAC inputs, using their means, or setting them to zero.

The sampler retains A_i as latent variables. Their own ATAC likelihood has $a = b = 0$, while their RNA-child correction remains active. This allows RNA evidence to inform an unobserved ATAC state and propagates its uncertainty back to RNA. It does not pretend that an ATAC measurement was observed. Conversely, for an ATAC-only row, integrating all unobserved RNA loadings gives $\int p(R_i | A_i, X_i) dR_i = 1$. At fixed parameters they provide no additional marginal ATAC likelihood, although sampling them can affect individual conditional moves and their joint draw can be used for prediction.

The accompanying example includes paired, RNA-only and ATAC-only rows, arbitrary held-out entries, and deliberately reordered RNA rows. Finite fixed covariates must be provided for every union ID. NaNs in measurement matrices represent missing likelihood terms; NaNs are not passed to the conditional networks.

Some paired samples or suitable external identifying information are needed to learn the cross-modality relationship. Without overlap, many distinct couplings can yield the same observed modality marginals; shared side information alone does not generally remove this ambiguity. The prototype therefore requires some paired rows for initialization. Even with overlap, an unpaired population far outside the paired population's support requires extrapolation. Neither the derivation nor the example guarantees improved reconstruction for both modalities.

E Variational and score-based alternatives

E.1 A numerical augmented variational update

The coordinate solution in Section 6.5 already applies to all the scalar families. An implementable approximation would draw uncertain parents from the current variational distribution, hold these draws fixed within a coordinate step, evaluate A_h, B_h, C_h and $D_u^{\text{aug}}(v)$, and integrate each tilted Gaussian using adaptive quadrature. Include the separate atom mass, normalize all component masses together, and update both first and second moments from the same integrals.

Two approximations must be kept distinct: Monte Carlo error in the expected log prior, and quadrature error in the one-dimensional tilt. Replacing a tilted component by a moment-matched Gaussian is an additional variational-family restriction. A finite collection of quadrature nodes by itself is a numerical integration device, not a continuous posterior density with a valid differential entropy. One cannot claim an ELBO for a node-based update while substituting Gaussian entropy without specifying this extra approximation.

A practical alternative is to explicitly restrict each continuous variational component to a Gaussian,

$$q_u(dv, h) = r_{uh} \mathcal{N}(dv; m_{uh}, s_{uh}^2) \quad (h \text{ continuous}),$$

with a separate atom when present. Evaluate the augmented ELBO by enumerating the target's component label, reparameterizing its Gaussian draw, and using Monte Carlo samples for other nodes. Its exact per-node entropy is

$$\mathcal{H}(q_u) = - \sum_h r_{uh} \log r_{uh} + \sum_{h \text{ continuous}} \frac{r_{uh}}{2} \log(2\pi e s_{uh}^2).$$

This is entropy on the *augmented* space, not the entropy of a collapsed overlapping Gaussian mixture. Full label enumeration across all uncertain parents is generally exponential; score-function estimators with variance reduction or structured approximations may be needed for their discrete variables. Relaxing a point mass to a continuous variable changes the approximation and requires separate validation. Fixed-parameter HMM variational block updates remain as derived in Section 9.

Mean-field VI can be much faster than long MCMC runs but can underestimate dependence and uncertainty. A row-structured $q(A_i, R_i)$, or a row-wise amortized inference model, can retain more cross-modality dependence at additional cost. There is no conjugate universal EBNM-only update for arbitrary neural conditional priors. The present code implements the sampling reference and exact within-component scores, not these proposed VI algorithms.

E.2 The exact conditional score is already available

For a fixed continuous target component h , the log conditional is

$$\log p(v, h \mid \text{rest}) = C - \frac{1}{2}av^2 + bv + \log w_{uh}(P_u) + \log \varphi(v; \mu_{uh}(P_u), d_{uh}^2(P_u)) + D_u(v).$$

Because the target's own parents are fixed during this coordinate,

$$\partial_v \log p(v, h \mid \text{rest}) = -av + b - \frac{v - \mu_{uh}(P_u)}{d_{uh}^2(P_u)} + D'_u(v). \quad (\text{E.1})$$

The implementation exposes this derivative by automatic differentiation of the normalized mixture model and verifies it against centered finite differences. No score-matching training is required to obtain it. For a Gaussian child with $e = V_c - \mu_c(v)$, its contribution is

$$\partial_v \log w_c(v) + \frac{e}{d_c^2(v)} \mu'_c(v) + \left(\frac{e^2}{d_c^2(v)} - 1 \right) \partial_v \log d_c(v). \quad (\text{E.2})$$

The last two terms are essential for EMDN. These derivatives are defined almost everywhere for the package’s ReLU networks; at a kink the software uses its automatic-differentiation convention.

E.3 What learning a score would and would not solve

Classical score matching fits continuous density scores without requiring the density’s normalization constant, under its regularity assumptions [2]. Here it could amortize repeated evaluations of (E.1) across contexts, using simulations or reference posterior draws. But component-specific continuous scores do not identify the relative component masses, and in particular do not determine an exact spike probability. The model still needs explicit discrete probabilities and transitions between the spike and continuous states. Training on observed noisy values also does not directly learn a latent loading prior without accounting for the likelihood.

There is a second obstacle for hierarchical priors. Suppose a conditional energy defines

$$g_c(v \mid P) = \exp f_c(v, P) / Z_c(P).$$

Its own value score $\partial_v f_c(v, P)$ omits $Z_c(P)$, but updating a parent requires

$$\partial_P \log g_c(v \mid P) = \partial_P f_c(v, P) - \partial_P \log Z_c(P).$$

The omitted normalizer generally depends on the parents. Learning conditional value scores alone therefore does not supply all the child messages needed for a coherent joint update. The normalized mixture parameterization avoids this obstacle.

A learned, evaluable proposal can still accelerate the reference sampler while retaining the target. For example, within one continuous component, propose

$$v' \sim \mathcal{N}\left(v + \frac{\epsilon^2}{2} s_\phi(v, \text{context}), \epsilon^2\right).$$

For a fixed proposal network, accept using

$$\alpha = \min \left\{ 1, \frac{p(v', h \mid \text{rest}) q_\phi(v \mid v', \text{context})}{p(v, h \mid \text{rest}) q_\phi(v' \mid v, \text{context})} \right\}. \quad (\text{E.3})$$

The full ratio is needed; the child-only cancellation in (C.5) applies specifically to q_0 . Interleave these continuous moves with valid joint value/component moves to reach other labels and the spike. An unevaluable implicit proposal cannot simply be inserted into this acceptance ratio. Adaptation must stop before ordinary fixed-kernel posterior collection, or meet a separately justified adaptive-MCMC condition.

The recommended sequence is therefore: validate the normalized joint target and component transitions; measure mixing and computational cost; then compare structured VI or learned proposals with the reference on small problems and held-out data. Score learning is a possible acceleration, not a substitute for the missing hierarchical terms.

F Verification and package correspondence

The formulas were checked algebraically by expansion and by the limiting cases listed below. The original companion numerical script compares small finite-state and one-dimensional calculations independently of the implementation:

1. No children and fixed parents recover ordinary CGB posterior moments.

2. A one-child tilted CGB conditional agrees with direct numerical integration of its spike and slab joint likelihood.
3. A binary two-node full conditional agrees with enumeration of all four joint states. Its mean-field update agrees with direct coordinate maximization and increases the same finite-state ELBO.
4. The EBNM-proposal Metropolis kernels satisfy detailed balance, and a complete two-coordinate sweep preserves the enumerated binary joint posterior.
5. The augmented EMDN update agrees with direct integration of the expected joint log density when uncertain parents change weights, means and variances, and an EMDN child contributes a downstream term.
6. The positive-slab evidence in (8.11) agrees with direct convolution, and the example’s RNA block mapping agrees with its code.

These checks validate the local identities, not convergence or empirical performance of a finite joint inference run. The associated numerical results are reported in the companion file `atac_rna_derivation_checks.txt`. The companion script `verify_atac_rna_derivation.py` reproduces these checks using Python, NumPy and SciPy. It is a verification script for this document, not a factorization solver.

The additional script `verify_mixture_family_derivation.py` independently checks the Gaussian, exponential and atomic identities in Section 7. It compares normalizers and moments with numerical integration, tests missing-likelihood and reflected-exponential cases, checks the augmented Metropolis ratio across component types, verifies exponential variational coordinates with uncertain parents, and verifies the autoregressive vector-mixture identity in Section 14.2. Results are recorded in `mixture_family_derivation_checks.txt`. These checks concern mathematical identities; they do not implement an exponential-mixture factorization solver.

The inspected source locations are relative to the repository root:

- `src/cebmf_torch/cebnm/cov_gb_prior.py`
Gate-only covariate dependence, signed global Gaussian slab, local fitting criterion and penalty.
- `src/cebmf_torch/cebmf/cebmf.py`
Loading and factor pseudo-data, posterior moments, covariate construction, local KL accounting and noise updates.
- `src/cebmf_torch/cebnm/hmm.py`
Unit-spaced HMM normal-means model, state mixtures, signed/one-sided variants and smoothing.
- `src/cebmf_torch/utils/standard_scaler.py`
Standardization recomputed from the current covariate matrix.
- `src/cebmf_torch/priors/learned.py`
Learned registry names and the common moment-returning adapter. The CASH, LCASH, sharp CGB and EMDN solver modules define the conditional families mapped in Section 6.4.
- `docs/source/ordered_loading_priors.rst` and `docs/source/hmm_priors.rst`
Current user-facing model semantics.

The local CGB solver’s moment-based slab-variance estimator is not being identified with the joint latent-variable M-step (10.2). This note defines the latter independently. The production cEBMF update is unchanged; the new experimental implementation is described in Section C. Its tests compare every scalar parent–child family pairing against the complete joint Metropolis ratio, numerical quadrature, native network outputs, finite-difference scores, enumerated HMM path probabilities, and fixed-parameter HMM posterior moments. Whole-solver checks exercise all eight families with missing modality rows and missing entries, fixed pre-processing, parameter freezing, and joint reconstruction moments. The executable tests are `tests/test_joint_conditional.py` and `tests/test_joint_atac_rna.py`. These are correctness tests, not a convergence certificate or a benchmark against competing methods.

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